

Strategic Underreporting Under Conditional Access: Evidence from Deep-Sea Mining

Andrés de Loera*

Harvard University

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Abstract

The International Seabed Authority (ISA) requires deep-sea mining contractors operating in international waters beyond national jurisdiction to conduct biological surveys of their prospectively-mined tracts, creating a public good as a byproduct of private exploration. However, contractors face a principal-agent problem: they have incentives to underinvest in species detection if discoveries could create obstacles to mining approval. I test for this “taxonomic deflation” using over 160,000 biological samples from the ISA’s DeepData database, comparing taxonomic reporting detail in prospectively-mined tracts versus non-contract areas. I show that samples from non-contract areas are significantly more likely to report species-level identification, a pattern that persists after controlling for contractor identity, year, and environmental variables. A counterfactual exercise suggests that tens of thousands of additional observations would include species identifications absent the observed reporting gap. I discuss the ISA’s exploration requirement as a second-best solution to the deep sea’s public goods problem and consider policy options for mitigating the resulting incentive distortions.

1 Introduction

Can the imprecision of deep sea mining data be explained by scientific difficulties alone? Biological exploration of the deep sea is genuinely difficult. Depths exceeding four kilometers require specialized remotely operated vehicles, and recovered specimens are frequently damaged in collection or chemically altered by fixation in ways that obscure diagnostic features; the expertise to identify them to species is scarce (Frutos et al. 2022). Much identification must in any case be made from seafloor imagery rather than physical specimens,

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which caps the resolution achievable (Horton et al. 2021), and even intact material often cannot be resolved by morphology alone (Kürzel et al. 2022). Yet the parties responsible for most of this exploration are not entirely impartial: they are mining contractors seeking permits to extract nodules from the seafloor, whose commercial interests do not necessarily align with scientific thoroughness. The Clarion-Clipperton Zone (CCZ), the most active target for deep-sea mining, is estimated to harbor over 7,000 metazoan species, the vast majority unknown to science (Rabone, Wiethase, et al. 2023), and recent surveys continue to find that most newly collected specimens belong to previously undescribed taxa (Jażdżewska et al. 2025). In such conditions, coarse identification could reflect the limits of the discipline—or it could be driven by strategic incentives. A contractor whose sampling uncovers a species that triggers conservation review faces delays, regulatory redesigns, and potential permit denial. I argue that the observed imprecision of deep-sea mining data is partially explained by strategic choices, not by scientific difficulties alone.

This incentive structure is not unique to ocean governance. Wherever a regulator relies on a regulated party to produce the information that determines whether the regulator restricts its activities, the regulated party has reason to underproduce precision. Landowners harboring potentially endangered species report less and survey less under the Endangered Species Act (Brown and Shogren 1998; Lueck and Michael 2003); firms required to self-monitor pollutant emissions report values that diverge systematically from independently measured ones (de Marchi and Hamilton 2006); pharmaceutical sponsors whose trial results determine approval display patterns of selective and delayed reporting (Turner et al. 2008). What distinguishes the deep-sea mining setting empirically is that the same firm produces records under both regulatory regimes within the same field campaign, allowing the incentive to be cleanly measured rather than inferred.

Environmental impact assessment is the gating step in deciding whether, how, and where mining should be permitted, and the assessment is only as good as the biological exploration that feeds it. Unfortunately, no single entity has aligned incentives to fund that exploration: mining contractors gain nothing from species discoveries that do not serve their commercial interests, and third-party explorers do not bear the opportunity cost of delaying mining if they survey slowly. The ISA’s operative response has been to make mining contractors do the exploration themselves as a precondition for permit approval. The arrangement uses prospective mining rents as a carrot for biodiversity reporting, but it creates a possible perverse incentive: contractors have reason to underexplore precisely those tracts whose biology bears on their own permit. I ask whether contractors engage in “taxonomic deflation” on those prospectively-mined tracts, by reporting exploration results at lower taxonomic precision relative to the non-contract areas where the same firms also occasionally collect samples. Notably, this latitude is built into the regulatory architecture itself, as the binding rules compel environmental data collection but leave the *level* of taxonomic resolution to non-binding

technical guidance. Deflation thus requires no falsification and no rule-breaking; it operates on a margin of discretion the ISA has expressly left open.

The ISA requires mining contractors to submit biological data from their exploration areas to the DeepData database. I exploit the fact that the same contractors also occasionally submit data from areas outside their exploration contracts, including Areas of Particular Environmental Interest (APEIs) — collectively, non-contract areas — where no mining permit is at stake. APEIs are central to the comparison: they are large conservation blocks sited under the ISA’s explicit “representativity” criterion, designed to span the major environmental gradients of the CCZ. They are, by design, the closest available analogue to contract tracts.¹ This within-contractor, across-zone comparison holds fixed the inherent difficulties of deep-sea taxonomy and the reporting infrastructure, isolating the effect of the permit incentive on reporting precision.

I begin with a model in which a contractor jointly chooses two instruments: exploration intensity (how many samples to collect) and taxonomic precision (how specifically to identify the organisms recovered). The two instruments face asymmetric regulatory incentives. More sampling unambiguously helps the contractor by demonstrating survey compliance. Precision cuts the other way: identifying organisms to species level is what risks uncovering a find that triggers conservation review and jeopardizes the permit, so what precision is worth to the contractor is scientific rather than regulatory. Outside contract areas, where no permit is at stake, the tension disappears. The model yields two testable predictions: first, that contractors should sample prospectively-mined tracts more intensively than non-contract areas, but second that they should report those samples with less taxonomic precision.

I test the predictions on over 160,000 biological specimen records from the CCZ, submitted by 12 mining contractors to DeepData. The dependent variable is whether each record carries a species-level identification. I estimate linear probability models that incrementally add controls—a baseline comparison of zone types, then contractor fixed effects (absorbing cross-firm differences in scientific capacity), year fixed effects (absorbing improvements in identification methods over time), and environmental controls matched to each sample’s coordinates from Bio-ORACLE rasters. The headline finding is robust across all four specifications: non-contract-area samples are more likely to include a species identification than prospectively-mined-tract samples, and the gap survives increasingly demanding controls. As a robustness exercise, I progressively condition on taxonomy fixed effects from phylum through genus; the gap persists even when I compare organisms classified to the same genus, ruling out compositional differences in sampled taxa as an explanation.

I then exploit a structural feature of the dataset—a small but consistent share of biological samples are

¹It should be noted, however, that they are not perfectly representative. APEIs do differ from contract areas, but primarily in their concentration of polymetallic nodules (Washburn et al. 2021).

collected by one contractor inside *another* contractor’s polygon—to run two complementary robustness designs. A within-sampler comparison of own-tract vs. guest-tract reporting holds scientific capacity fixed at the firm level; a within-polygon comparison of owner-collected vs. guest-collected samples holds ecology fixed at the tract level. Both designs deliver coefficients consistent with the headline result.

To probe the channel through which deflation operates, I test the addition of cruise-leg fixed effects and identifier-name fixed effects to the main specification. Cruise-leg effects absorb about half of the documented gap, implying roughly half of the deflation reflects between-expedition selection, with different cruises sampling different zones with different teams. Identifier-name effects absorb essentially the rest: the same individual taxonomist produces statistically indistinguishable species-reporting rates across contract and non-contract samples. The deflation operates primarily through the *assignment* of identification effort. Since scientific analysis of exploration cruises is depends on contractors sponsoring taxonomic effort, this suggests contractors use less-precise identifiers when dealing with permit-relevant samples. I also investigate other related mechanisms, such as the assignment to scientific institutions and the volume of samples, which do not change the headline result.

Next I run a counterfactual exercise that simulates species reporting as if every observation were reported with non-contract-area-level precision, holding exploration intensity and all other covariates fixed. The exercise decomposes the missing species-level identifications in the dataset into two parts: a portion attributable to strategic incentives (the deflation wedge) and a portion attributable to the genuine limits of deep-sea taxonomy. Even under the most conservative coefficient, strategic incentives account for roughly seven percent of missing species identifications—on the order of 8,000 specimens under the conservative bound and as many as 37,000 under the largest coefficient, all of which would carry a species name absent taxonomic deflation. A companion exercise turns to the underexploration outside contract areas. APEIs, the conservation blocks the ISA designed as a representative biodiversity benchmark, are surveyed more than 600 times less densely than the tracts contractors intend to mine; had they been sampled at mined-tract density, the few hundred specimens they have produced would instead run to the hundreds of thousands, carrying on the order of 95,000 additional species-level identifications. While these counterfactual thought experiments are large extrapolations, they suggest strategic incentives have a large role to play in explaining the lack of biodiversity data from the deep sea.

I close with policy options, organized around the two wedges the model identifies—though, as the model also shows, on a contract area neither instrument is confined to a single margin. Section 2 presents the institutional background, Section 3 the model, Section 4 the empirical methods, Section 5 the results (with the mechanism and assignment-channel analyses), Section 6 the interpretation, Section 7 policy responses,

and Section 8 concludes.

My analysis draws on three strands of the economics literature. The first is the literature on strategic manipulation of self-reported environmental data. Firms' self-reported environmental data routinely diverge from independently measured values (de Marchi and Hamilton 2006), and third-party auditors can be compromised when their compensation depends on the regulated firm (Duflo et al. 2013). Regulated agents respond dynamically to enforcement, adapting their evasion strategies as they learn the inspection regime (Gonzalez-Lira and Mobarak 2021), and regulatory pressure can paradoxically reduce the quantity and quality of voluntary disclosures (Zakota 2025). In some cases agents manipulate the monitoring process itself rather than the data it produces.² Mandatory disclosure requirements can meaningfully improve environmental outcomes (Benneer and Olmstead 2008; Doshi et al. 2013), though firms also selectively disclose favorable information while withholding unfavorable results (Lyon and Maxwell 2011). The form of misreporting I document operates through a different margin: contractors do not fabricate coarse identifications; they rationally forgo the precise ones.

The second strand concerns perverse incentives in species-protection regulation. Brown and Shogren (1998) and Langpap et al. (2018) survey the economics of the Endangered Species Act (ESA), emphasizing the tension between conservation benefits that accrue nationally and costs that fall on private actors harboring species. The tension generates strategic responses: landowners near known endangered-species colonies preemptively destroy potential habitat to avoid triggering ESA protections (Lueck and Michael 2003), and ESA policy gives landowners no incentive to cooperate with species-information collection on their land (Polasky and Doremus 1998). Institutional design can mitigate these incentives: safe-harbor agreements reduce preemptive habitat destruction (Byl 2019). The mechanism in my setting is structurally similar—an agent with a valuable asset manipulates the conditions that would trigger conservation oversight—but the manipulation is informational rather than physical. Landowners destroy habitat; mining contractors suppress taxonomic precision.

The third strand is the literature on regulatory enforcement and compliance under asymmetric information (Harrington 1988; Heyes and Rickman 1999). The ISA's exploration requirement uses private rents as a carrot for public-good provision, and the taxonomic deflation I document is precisely the kind of shirking on uncontracted margins that such arrangements invite.

Across these strands, I develop three conceptual contributions. First, I distinguish *strategic imprecision*—the rational underproduction of information that the agent is required to produce—from the more familiar

²Mu et al. (2021) show that local governments strategically shut down air-pollution monitors on days when air quality is expected to deteriorate. Kim and Lyon (2011) find that firms in the DOE's Voluntary Greenhouse Gas Registry reported emissions reductions even as their actual emissions increased.

problems of falsification, evasion, and selective disclosure. Where misreporting requires the agent to assert a false fact, strategic imprecision requires only that the agent stop short of a true one. That is, the channel I document does not require any deception; it requires only an institutional setting in which silence is cheaper than precision. I also coin a specific case of strategic imprecision, *taxonomic deflation*, which is the inverse of taxonomic inflation (Isaac et al. 2004). Second, I provide a tractable model in which exploration intensity and reporting precision are chosen jointly, showing that the prospect of a mining permit is on its own enough to raise sampling and depress precision at the same time—a joint pattern that differences in capacity or cost would not produce—and that the resulting welfare loss has two distinct sources, each answering to a different policy instrument. Third, I identify the *channel* through which strategic imprecision operates in this setting: not direct misreporting by individual scientists, but allocation of identification effort across samples by the firms that employ them. This relocates the empirically relevant policy lever from individual-reporting standards to the institutional design of sample assignment, with implications for any setting in which the regulated firm retains discretion over who processes the samples used to evaluate it.

2 Background

2.1 The Deep Sea as a Commons

The resources of the ocean seabed beyond national jurisdiction are designated “the common heritage of mankind” under the United Nations Convention on the Law of the Sea (UNCLOS). Those resources include critical minerals—cobalt, nickel, manganese, lithium—but also biodiversity. The minerals have immediate commercial value and could prove useful for decarbonization, but deep-seabed mining is widely expected to destroy valuable biodiversity in the process (Niner et al. 2018; Levin et al. 2020). The concern is not speculative. Jones et al. (2025) revisited experimental mining tracks in the CCZ forty-four years after disturbance and found that large sessile fauna had barely recovered; the first industrial-scale mining trial documented a 37% reduction in macrofaunal abundance and a 32% reduction in species richness in directly impacted areas (Stewart et al. 2026). The baseline against which those losses must be measured is, at present, barely characterized: less than 0.001% of the deep seabed has been visually observed (Bell et al. 2025), and Rabone, Wiethase, et al. (2023) estimate that the CCZ alone likely harbors over 7,000 metazoan species, the vast majority unknown to science. Bonifácio et al. (2021) document substantial morphological and genetic diversity among deep-sea scale-worms (Polynoidae); Jażdżewska et al. (2025) catalog the CCZ’s Amphipoda and find that most discovered species were previously unknown. So before any decision about mining can be made on a scientifically grounded basis, the exploration has to happen. Yet exploration has historically fallen

short of describing the rich variety of life in the deep sea. Species discovery and identification are rarely funded (Glover et al. 2018; Britz et al. 2020)—a shortfall that Glover et al. (2018) calls the “historic failure of taxonomy on the high seas.”

2.2 The ISA’s Exploration Requirement

The ISA manages the deep seabed and has issued exploration contracts entitling country-backed contractors to survey a specific tract (Blanchard et al. 2023). The ISA officially terms these “exploration areas.” Because what drives the incentives I study is that a single contractor holds and expects to mine them, I call them *contract areas* throughout—using *prospectively-mined tracts* interchangeably where the mining prospect is the salient feature—to contrast them with conserved areas. The eventual goal of an exploration contract is conversion to an exploitation contract, but the present requirement is that the firm intending to mine an area must first collect environmental and biological data about it. The straightforward justification: the data are inputs to the ISA’s decision over whether mining should be allowed in a particular location. The deeper rationale: the requirement forces prospective miners to produce a public good—non-rival, non-excludable information about the marine genetic resources of the deep sea—that the ISA needs in order to balance mining against conservation (Jones et al. 2019).

A central deliberation over whether deep-sea mining should be permitted at all turns on undiscovered species and the precautionary principle (Jaekel 2017). If firms must fund exploration before exploitation, mining cannot destroy a valuable ecosystem or genetic resource before scientists can characterize it. Deep-sea organisms have become increasingly prominent in marine bioprospecting, with growing numbers of patent filings tied to deep-sea genetic material (Zhivkoplis et al. 2024; Krusberg et al. 2024). If the seabed might harbor valuable biological resources, there is quasi-option value to delaying mining in favor of exploration (Arrow and Fisher 1974; Henry 1974), particularly given evidence that mining impacts persist for decades and that recovery is at best partial (Jones et al. 2025). The catch is that no one outside the ISA has aligned incentives to fund that exploration on the timeline mining requires. Pharmaceutical companies that would eventually benefit from deep-sea genetic discoveries cannot reliably secure intellectual property rights over them, so they have little reason to bioprospect now. So the ISA’s exploration requirement is a clever workaround: it forces the contractor to bear both the cost of exploration and the opportunity cost of delaying mining, in exchange for the right to mine eventually. The obvious alternatives—granting the contractor exclusive rights over biological resources as well as minerals, or pausing mining indefinitely until a scientific body deems the deep sea “sufficiently explored”—are legally infeasible and each creates its own tension between exploration and conservation.

What that requirement actually compels—and what it leaves to the contractor’s discretion—is worth setting out precisely, because the distance between the two is where taxonomic deflation operates. The binding instruments are the Authority’s exploration regulations and the standard contract clauses they incorporate: each contractor “shall” gather environmental baseline data, establish baselines, run a monitoring program, and report the results annually, and its application must describe a program to assess the “impact on biodiversity” of the proposed activities (International Seabed Authority 2013). However, the binding regulations stop there: They never enumerate which taxa must be characterized, never specify a taxonomic level, and twice route the question of *how* to recommendations issued by the Legal and Technical Commission—guidance that contractors are directed merely to “take into account” (International Seabed Authority 2013). Those recommendations, which are the technical rules contractors actually follow, are written almost entirely in the register of “should” rather than “shall” (International Seabed Authority 2020a). They explicitly endorse “taxonomy by numbers”—numbered morphospecies backed by voucher specimens—as “a good basis for baseline studies,” treat image-based megafaunal surveys as resolving only organisms larger than two centimeters and requiring physical specimens to confirm any species identification, and let a contractor petition for exemption from any specific recommendation it finds infeasible (International Seabed Authority 2020a). The Authority itself has recognized that it “can only recommend” taxonomic approaches and that consistent, enforceable standards have yet to be set (International Seabed Authority 2020b). The upshot is that a contractor can satisfy every binding obligation—survey the tract, archive its specimens, file its annual reports—while reporting what it recovers only to morphotype or genus (or even coarser). Species-level identification is encouraged, but not required. This institutional background sets the stage for the taxonomic discretion that the model in the next section formalizes.

2.3 Taxonomic Deflation and the Comparison Strategy

As discussed, the ISA’s solution to underexploration comes with a possible downside: it can incentivize the contractor to identify what it recovers only coarsely. Because the contractor’s returns come only from eventually mining the tract, it has strong reason to avoid discovering anything that might lead to its permit being denied, such as rare species, novel taxa, charismatic fauna. Species-level identification also matters for conservation status, and for the regulatory review and delay that such status can invite (Delić et al. 2017). This incentive could be partially addressed by mandating high standards for the quality and thoroughness of reporting, by requiring third-party explorers, or by imposing other procedural safeguards. But the incentive can still operate through uncontracted margins: the contractor can hire different teams of scientists, use different equipment, or shade categorization decisions when the underlying classification is uncertain. In

particular, the contractor can attempt *taxonomic deflation*—the flip side of “taxonomic inflation,” in which subpopulations are erroneously categorized as independent species (Isaac et al. 2004). Taxonomic inflation has been used in service of conservation, by making subpopulations eligible for endangered species protection (Ghezelayagh et al. 2025). Taxonomic deflation does the opposite: it undermines conservation by failing to identify distinct species that should be considered separately for protection. Both manipulations exploit the same underlying fact—the species concept is hard to define precisely, and that difficulty leaves room for parties with their own goals to push the boundary either way (George and Mayden 2005). The species is nonetheless the fundamental unit of biodiversity, so “deflation” to the phylum or class necessarily understates a substantial amount of biodiversity (Claridge et al. 1997).

All biological and environmental samples collected under an exploration contract must be reported to the ISA in a prescribed annual-report format (International Seabed Authority 2015) and are eventually made public through the DeepData database. That format standardizes the *fields* a contractor populates, but it imposes no floor on the taxonomic *resolution* those fields must reach. The quality of the resulting reports varies considerably. Rabone, Horton, et al. (2023) document persistent issues, including duplication of entries, inconsistent or absent identifiers, and substantial data-quality problems in the taxonomy variables. Even absent strategic underreporting, deep-sea biological data are often taxonomically coarse (Costello and Chaudhary 2017; Kennedy and Rotjan 2023), and the ISA has been markedly less transparent than other international resource managers (Ardron 2018). Just 29,445 of the 160,967 CCZ records in DeepData (18%) carry a species-level identification at all. Therefore pointing out the absolute lack of taxonomic information in DeepData is not enough: the data from prospectively-mined tracts has to be compared against an appropriate counterfactual.

That counterfactual is the same contractors’ submissions from areas outside their own tracts—Areas of Particular Environmental Interest (APEIs) and areas outside any formal zone designation—which I refer to collectively as non-contract areas. Every sample in DeepData, including those from APEIs and outside areas, is submitted by a mining contractor as part of an exploration program. The comparison is therefore within the same reporting entities, holding fixed the inherent difficulties of deep-sea exploration, the DeepData reporting infrastructure, and each contractor’s scientific methodology and institutional partnerships. If the same contractors’ data show systematically less taxonomic detail in contract areas than in non-contract areas, that pattern is consistent with taxonomic deflation. The APEI comparison is conceptually the cleanest—these zones are explicitly designated to be free of mining, so biodiversity reporting there carries no permit consequences. Regression results restricted to the APEI-only comparison are reported in the Appendix and are consistent with the main analysis. Figure 1 shows the administrative boundaries of the CCZ, with

contractor tracts (colored) and APEIs (gray) labeled. The spatial distribution of biological samples in the DeepData database is shown in Appendix Figure 10.

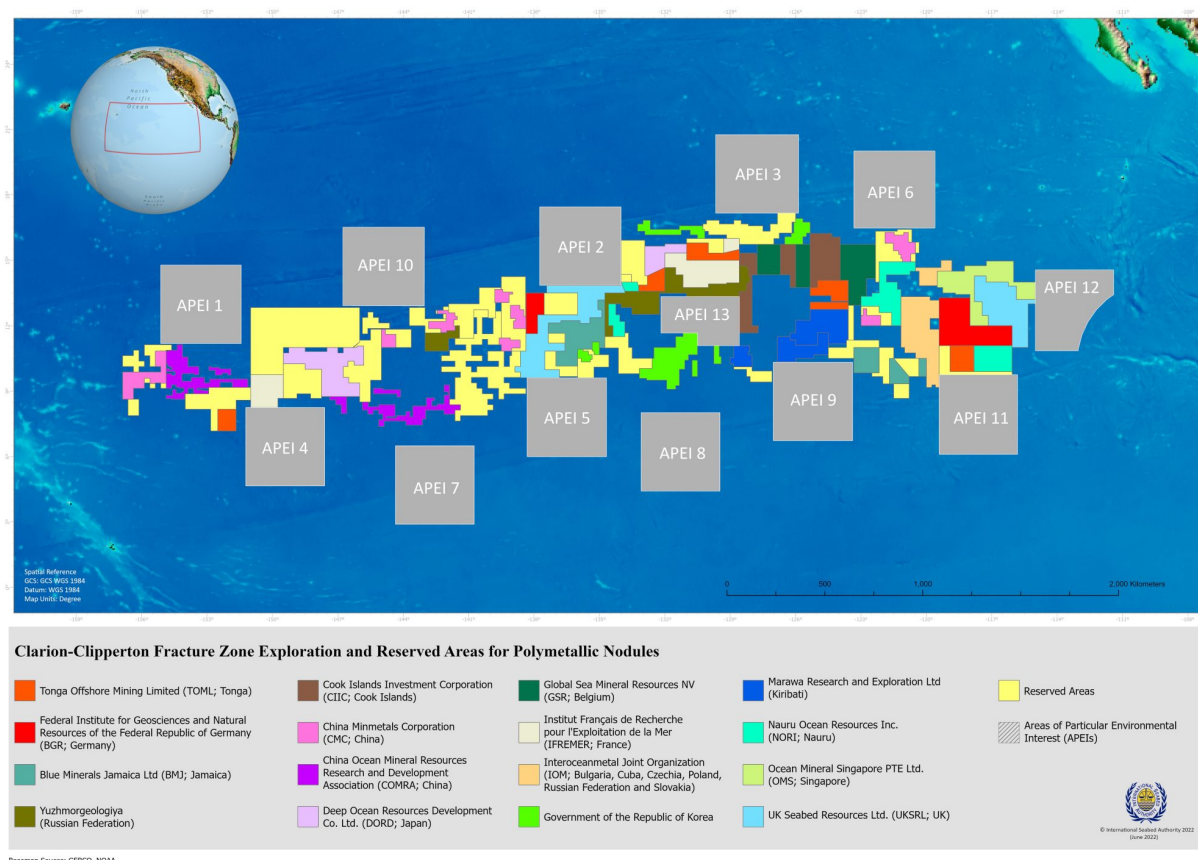


Figure 1: Contractor exploration areas (colored) and Areas of Particular Environmental Interest (gray) in the Clarion-Clipperton Zone. Source: International Seabed Authority.

3 Model

Before turning to the empirics, I develop a model that clarifies the mechanism behind the empirical strategy. The model shows that the observed pattern of taxonomic deflation follows from rational contractor behavior under the ISA's regulatory structure. Two predictions emerge, each corresponding directly to one set of empirical findings below: contractors sample prospectively-mined tracts more intensively than non-contract areas, but report those samples with less taxonomic precision.

The core idea is that exploration intensity and taxonomic precision are distinct instruments facing asymmetric regulatory incentives. More sampling unambiguously raises a contractor's odds of regulatory approval by demonstrating survey compliance. Precision cuts the other way: carrying an organism to species level can

produce a finding, such as a rare or valued species, that triggers conservation review and jeopardizes the mining permit. Precision may still be worth something to the contractor through other channels, such as scientific returns or some regulatory minimums, of course. Outside contract areas, where no permit is at stake, only the scientific return operates, and contractors have no reason to withhold precise identifications.

3.1 Setup

Consider a mining contractor that jointly chooses two instruments: exploration intensity $e \geq 0$, capturing the volume of biological sampling it conducts, and taxonomic precision $\tau \in [0, 1]$, capturing how specifically it identifies the organisms recovered.³ Although any single specimen is identified to a discrete taxonomic rank, treating τ as continuous is natural at the level of a survey: it represents the *share* of recovered organisms the contractor carries to species level, which varies smoothly with the identification effort it allocates. The two instruments are conceptually and empirically distinct—a contractor can collect many samples while reporting all observations only at the phylum level, or conduct a small but meticulous survey. The DeepData database encodes exactly this distinction: it records both the number of biological samples submitted and the taxonomic level at which each is identified.

The regulator observes both instruments and makes an approval decision. I write the probability of approval as a product of two forces:

$$\pi(e, \tau) = \underbrace{A(e)}_{\text{compliance}} \cdot \underbrace{S(\tau)}_{\text{survival}} \quad (1)$$

$A(e)$ is the probability that the survey satisfies the Authority’s requirements, and it rises with sampling: this is the compliance benefit of exploration. $S(\tau)$ is the probability that nothing the contractor identifies blocks the permit, and it falls with precision: this is the discovery hazard. The product form implicitly assumes that the two are independent hurdles—the survey must both satisfy the regulator and turn up nothing disqualifying.

A survey has some value to the contractor apart from the permit, although it may be quite small. I write that value as $G(e) + H(\tau)$: G is the value of a survey’s *coverage* and H the value of its *resolution*, each already net of what it costs to produce. In this model, the contractor pushes each instrument to the point where that instrument’s own marginal benefit meets its own marginal cost, and does so in each zone separately. This is a natural description of the setting, as sampling happens at sea and identification happens onshore, months

³In practice the activity itself is likely contracted out to scientific partners, so these choices are stand-ins for the contract terms and funding dedicated to this endeavor.

later and usually by different people.

Suppose the mining permit is worth $V > 0$. Letting $M = 1$ denote a prospectively-mined tract, the contractor's expected payoff is

$$\Pi(e, \tau; M) = \mathbb{K}\{M\} \cdot A(e) S(\tau) \cdot V + G(e) + H(\tau). \quad (2)$$

The two zones share one objective and differ only by the indicator on the permit term, so setting $V = 0$ returns the contract problem exactly to the non-contract problem. I impose three restrictions on functional forms:

$$\textbf{(A1)} \quad A' > 0, \quad A \in [0, 1)$$

$$\textbf{(A2)} \quad S' < 0, \quad S \in (0, 1], \quad S' \text{ bounded}$$

$$\textbf{(A3)} \quad G, H \text{ strictly concave, continuously differentiable, with interior peaks,}$$

and $G'(0) = H'(0) = +\infty$.

(A1) assumes that more sampling leads to a higher probability of compliance, at a rate that is everywhere strictly positive.⁴ (A2) says that precision raises discovery risk, that no level of precision makes denial certain, and that the risk a marginal identification adds is finite.⁵ (A3) says coverage and resolution each pay off with diminishing returns, are eventually not worth what they cost, and that the first unit of each is always worth buying.⁶ These minimal assumptions do not impose a specific functional form, but just restrict the *direction* of each force. I do not assume the contractor's objective is concave on a contract area—in general it is not, since a discovery hazard is naturally convex—nor that its optimum is unique.⁷

However, for concreteness and to produce closed forms in later sections, consider the following:

$$A(e) = 1 - e^{-\alpha e}, \quad S(\tau) = e^{-\phi \tau}, \quad G(e) = b_e \sqrt{e} - \kappa e^2, \quad H(\tau) = b_\tau \sqrt{\tau} - \lambda \tau^2,$$

⁴The strictness matters: a contractor facing a locally flat compliance return at the science optimum would have no reason to sample past it, and Prediction 1 would survive only in weak form.

⁵Bounding S' , together with the Inada conditions in (A3), is what keeps the contractor's own optimum interior: as precision approaches zero the marginal scientific return to resolution grows without bound while the marginal hazard stays finite, so the contractor never abandons identification altogether.

⁶Differentiability is required on the whole interior rather than only at the peaks, because the welfare wedges of Section 3.4 are evaluated at the contractor's own optimum—which Predictions 1 and 2 place strictly away from those peaks.

⁷A maximizer exists: $A(e)S(\tau)V \leq V$ is bounded while (A3) sends $G(e) \rightarrow -\infty$ at least linearly, so the objective is coercive in e and the problem may be restricted to a compact set on which it is continuous.

with $\alpha, b_e, b_\tau, \kappa, \lambda > 0$ and $b_\tau < 4\lambda$, which places the resolution peak inside $[0, 1]$. Here α is the rate at which sampling buys compliance, b_e and b_τ are the scientific returns to coverage and resolution, κ and λ are their costs, and $\phi > 0$ is the regulatory bite of a species discovery (the rate at which precision erodes the odds of approval). The cost parameters sit inside the science functions, since G and H are what coverage and resolution are worth net of what they cost. The central predictions hold without these specific functional forms, however.

3.2 Optimal Choices in Non-Contract Areas

In non-contract areas there is no permit at stake, and the contractor faces no regulatory penalty for precise identification. Its problem collapses to

$$\Pi_A(e, \tau) = G(e) + H(\tau), \quad (3)$$

which is additively separable and strictly concave in each argument, so it has a unique solution, obtained by setting each instrument at the peak of its own return: $G'(e_A) = 0$ and $H'(\tau_A) = 0$. Off contract, in other words, the contractor runs the survey that its scientific incentives alone would justify. The two instruments decouple completely: e_A and τ_A are constants, neither depending on the other, and neither depending on anything about the permit. Under the illustrative forms,

$$e_A = \left(\frac{b_e}{4\kappa}\right)^{2/3}, \quad \tau_A = \left(\frac{b_\tau}{4\lambda}\right)^{2/3}. \quad (4)$$

3.3 The Central Predictions

Two facts about (2) drive both predictions. First, on a contract area sampling and precision are substitutes: the more the contractor has already sampled, the less an additional identification is worth to it. Formally, for $e' \geq e$ and $\tau' \leq \tau$,

$$\Pi(e', \tau') + \Pi(e, \tau) - \Pi(e', \tau) - \Pi(e, \tau') = \mathbb{K}\{M\} V [A(e') - A(e)] [S(\tau') - S(\tau)] \geq 0.$$

Second, the permit raises the value of sampling and lowers the value of precision: $\Pi(\cdot; 1) - \Pi(\cdot; 0) = A(e)S(\tau)V$ is increasing in e by (A1) and decreasing in τ by (A2). Both facts use only the monotonicity of A and S , so they do not depend on the specific functional forms, only on the minimal assumptions imposed above.

These are the conditions under which the theory of monotone comparative statics applies, provided the precision dimension is read in reverse, with the contractor choosing *coarseness* rather than precision (Topkis 1978; Milgrom and Shannon 1994).⁸ Read that way the contractor's two choices are complements, since the more it has sampled the more it wants to deflate, and the permit raises the return to both at once. The theorem then says that raising the parameter, here the contract indicator itself, raises the contractor's optimal choice of each: it samples more and identifies more coarsely. Because the non-contract optimum is a single point, this holds for *every* optimal choice the contractor might make on a contract area, not only the extreme ones.⁹ Hence $e_M \geq e_A$ and $\tau_M \leq \tau_A$.

Strictness follows from evaluating Π_M at the benchmark, which (A3) places in the interior. Since $S(\tau) > 0$ for every τ by (A2), we get

$$\left. \frac{\partial \Pi_M}{\partial e} \right|_{e_A} = \underbrace{A'(e_A) S(\tau) V}_{> 0} + \underbrace{G'(e_A)}_{= 0} > 0 \quad \text{for any } \tau, \quad (5)$$

so e_A is not a critical point of the contract-area problem and no optimum sits there. That gives $e_M > e_A > 0$, so $A(e_M) > A(0) \geq 0$ strictly, A being strictly increasing, and therefore

$$\left. \frac{\partial \Pi_M}{\partial \tau} \right|_{\tau_A} = \underbrace{A(e_M) S'(\tau_A) V}_{< 0} + \underbrace{H'(\tau_A)}_{= 0} < 0. \quad (6)$$

Collecting these:

Prediction 1: More Exploration on Prospectively-Mined Tracts. *For every $V > 0$, the contractor explores a prospectively-mined tract more intensively than a non-contract area: $e_M > e_A$.* Sampling buys compliance, and compliance is worth something only where a permit is at stake. Off contract the contractor samples only as much as the science warrants; on contract it samples more, at any permit value, because (5) is strictly positive whenever $V > 0$.

Prediction 2: Less Precision on Prospectively-Mined Tracts. *For every $V > 0$, the contractor reports a prospectively-mined tract less precisely than a non-contract area: $\tau_M < \tau_A$.* Off contract, precision carries no regulatory risk and the contractor identifies to the point where the scientific return to resolution is

⁸Formally, order the choice set by $(e, -\tau)$, which reverses the order on the second dimension and leaves the choice set a lattice. Under that order the two displayed facts are exactly supermodularity of Π and increasing differences in M , which are the theorem's hypotheses.

⁹The conclusion of the theorem is that the contract-side optimal set dominates the non-contract set in the strong set order: for any y in the former and x in the latter, the componentwise minimum of y and x lies in the latter. When the latter is the singleton $\{(e_A, -\tau_A)\}$, that minimum must equal $(e_A, -\tau_A)$ itself, which says precisely $y \geq (e_A, -\tau_A)$ — that is, $e_M \geq e_A$ and $\tau_M \leq \tau_A$.

exhausted. On contract, each additional identification carries the risk of a finding that blocks the permit, so the contractor stops short: by (6), it is already better off pulling back from the precision that science alone would warrant. What the regressions below estimate is the per-sample term τ , whose empirical counterpart is the share of submitted samples carrying a species-level identification, and I take up in Section 3.5 what $\hat{\beta}$ recovers.

The two predictions are not independent. Sampling and precision are substitutes on a contract area and independent off one: $\partial^2 \Pi / \partial e \partial \tau = \mathbb{1}\{M\} V A'(e) S'(\tau)$, negative when $M = 1$ and zero when $M = 0$. The hazard's bite is proportional to $A(e)$ —the compliance value the contractor's sampling has earned—so the more it has surveyed, the more an adverse discovery can cost it, and the stronger incentive it has to deflate.

3.4 Further Implications

The model generates additional comparative statics that, while not directly tested here, sharpen intuition about how the deflation mechanism operates.

Asset value and deflation intensity. More valuable tracts should draw more sampling and coarser identification, all else equal. e_M is nondecreasing and τ_M nonincreasing in V , by the same argument as the predictions with V in place of the contract indicator, since raising V raises the marginal value of sampling ($\partial^2 \Pi / \partial e \partial V = A' S > 0$) and lowers that of precision ($\partial^2 \Pi / \partial \tau \partial V = A S' \leq 0$). More commercially valuable deposits, those with greater nodule density or surveyed during periods of high commodity prices, should produce larger sample counts but coarser identifications within the same contractor. Testing this directly could be possible with estimates of nodule abundance or value that are not currently publicly available. However, even with this information, it is possible different contractors also differ in their non-permit benefits or costs, which makes this only a conditional prediction.

Regulatory bite. The deflation gap need not widen as the consequences of a species discovery grow more severe. Because a larger ϕ also deters sampling, and because sampling and precision are substitutes on a contract area, less sampling pushes precision back up. As written, the model does not sign the net effect, though the effect on the total number of species detections is unambiguously negative. In any case, testing the effect of regulatory bite directly would require variation in regulatory stringency, which is not available.

These behavioral implications have direct welfare consequences. Consider a social planner who values a precisely-identified species at $s > 0$ —capturing option value for conservation, ecosystem services, and future research—and who does not treat a discovery as a loss, since blocking mining redirects the tract to protection rather than necessarily destroying surplus:

$$W(e, \tau) = A(e) V + G(e) + H(\tau) + s \cdot e \cdot \tau. \quad (7)$$

The scientific term enters unweighted by the approval probability: the knowledge from a precisely-identified sample accrues to society whether or not the permit is ultimately granted, unlike the contractor's permit payoff, which is contingent on approval. The discovery hazard, meanwhile, enters the contractor's objective but not the planner's—society is not harmed by learning that a tract holds a rare species.¹⁰ Evaluating the planner's marginal valuations at the contractor's own optimum on a prospectively-mined tract gives two wedges, both strictly positive:

$$\left. \frac{\partial W}{\partial \tau} \right|_{(e_M, \tau_M)} = \underbrace{-A(e_M)S'(\tau_M)V}_{\text{hazard the planner does not face}} + \underbrace{s e_M}_{\text{unpriced information}}, \quad (8)$$

$$\left. \frac{\partial W}{\partial e} \right|_{(e_M, \tau_M)} = \underbrace{A'(e_M)V[1 - S(\tau_M)]}_{\text{hazard also deters sampling}} + \underbrace{s \tau_M}_{\text{unpriced information}}. \quad (9)$$

The planner wants more of both sampling and precision than the contractor supplies. The precision wedge rests on less than the exploration wedge does. Prediction 2 puts τ_M strictly below τ_A , and H is strictly concave with its peak at τ_A , so $H'(\tau_M) > 0$ and $\partial W / \partial \tau = H'(\tau_M) + s e_M > 0$ follows without any appeal to the contractor's first-order condition.¹¹ An exploration wedge is present in non-contract areas too, though it is a different object. There society's problem is $W_A(e, \tau) = G(e) + H(\tau) + s e \tau$, with no permit to grant, so evaluating at the contractor's own choice gives $\partial W_A / \partial e|_{e_A} = s \tau_A$, the unpriced information alone. Neither ϕ nor V appears. Off contract, then, the shortfall in sampling answers only to instruments that price discovery; reducing the regulatory bite does nothing there.

Two distinct sources of divergence appear, and they call for different remedies. The first is the permit risk ϕ , which the contractor faces and the planner does not. Reducing it—through safe-harbor provisions, sealed reporting, or independent review of contested identifications—can reduce the first term of each wedge (at costs discussed later). The second is the scientific value s , which society enjoys and the contractor does not internalize. Making the contractor internalize it—through prizes, or through a market in marine genetic

¹⁰Two features of (7) are normalizations here. Society is taken to value an approved tract at the same V the contractor does, and the compliance term is retained unweighted by S , which treats a permit blocked by a discovery as costless to society but a permit blocked by non-compliance as a loss. Weakening either changes the magnitude of the wedges, but not the sign of the precision wedge.

¹¹Both statements are local: they say the planner would want more of each instrument starting from the contractor's optimum, which is what the policy discussion draws on. How far the planner's own optimum lies from the contractor's is not determined here, for an interesting reason—the contractor treats sampling and precision as substitutes while the planner, who values the *product* $e\tau$, treats them as complements. As a result, no single ordering of the choice set covers both problems.

resources—can reduce the second (also at a cost). On a contract area both sources appear in both wedges, so neither instrument is confined to one margin there. Off contract, as above, only the second has anything to work on. I discuss these instruments further after the empirical analysis.

3.5 Mapping to the Empirical Design

The model maps naturally onto the regression design below. The contract-area indicator partitions observations into those where the permit is at stake ($M = 1$) and those where it is not ($M = 0$). Contractor fixed effects absorb firm-level variation in the returns and costs embedded in G and H , and in the compliance technology A , leaving the permit as the only thing that differs between a contractor’s own tracts and the areas it surveys elsewhere. Under this mapping, the estimated coefficient $\hat{\beta}$ on the contract-area indicator recovers the precision gap $\tau_M - \tau_A$ —the difference in species-reporting rates induced by the regulatory mechanism, holding contractor and year fixed. The counterfactual analysis below—which simulates non-contract-area reporting precision for every observation—corresponds to shutting down the $\phi > 0$ mechanism while holding all else fixed. The model also clarifies why sampling volume cannot explain the finding: Prediction 1 implies that prospectively-mined tracts have more samples, so any purely ecological or resource-constraints explanation would predict *better* species coverage there, not worse. The observed reversal is the signature of strategic taxonomic deflation.

4 Methods

I test for taxonomic deflation by examining the taxonomic resolution of data reported on DeepData. Each observation is a biological specimen record submitted by a contractor to the ISA. When reporting to DeepData, contractors can populate up to eight taxonomic levels: Kingdom, Phylum, Class, Order, Family, Subfamily, Genus, and Species. Far more commonly, records report only as far as the phylum or class. I focus on whether a record carries a species-level identification, since the species is the fundamental unit of biodiversity assessment (Claridge et al. 1997). Species-level identification is also a margin the ISA’s technical guidance leaves to contractor discretion: its recommendations treat numbered morphospecies and genus-level identifications as an acceptable baseline, so a contractor that stops short of a species determination remains fully within the standard (International Seabed Authority 2020a). Taxonomic practice even supplies a standardized open-nomenclature vocabulary—qualifiers such as “cf.” and “aff.”—for recording identifications that stop short of a confirmed species (Horton et al. 2021), so coarse reporting reflects accepted scientific convention rather than obvious neglect. The dependent variable is therefore binary: it equals one if the record includes a species name, zero otherwise. I compare species reporting in samples from prospectively-mined tracts to

those from non-contract areas, accounting for the range of confounders the dataset allows.

The dataset contains over 160,000 biological specimen records from the CCZ, submitted by 12 mining contractors to DeepData. The records span prospectively-mined tracts (allocated to specific contractors for mineral exploration), APEIs, and areas outside both zone types. Every record in the dataset, including those from non-contract areas, was submitted by a mining contractor; the dataset does not include samples from independent academic expeditions or other non-contractor sources. That fact is central to the empirical strategy—the comparison is between samples from different zone types reported by the same set of contractors, not between contractor data and independently-collected scientific data. I match each record to environmental data from Bio-ORACLE rasters (Assis et al. 2024) using its reported longitude and latitude, yielding a matched dataset of 160,967 observations, of which 150,104 have complete environmental covariates and enter the specifications that add environmental controls. Table 5 in the Appendix reports summary statistics by zone type, including the number of records, year range, and species-reporting rates.

I begin by showing the likelihood of species reporting in the raw data, then proceed to regression models that estimate the probability a specimen record includes a species identification. The main explanatory variable is whether the sample comes from a contractor-allocated exploration tract (a contract area). I present four specifications that incrementally add controls, showing how the estimated effect changes—and survives—as potential confounds are absorbed. I estimate a linear probability model (LPM) with heteroskedasticity robust standard errors: the object of interest is the average marginal effect of contract-area status on a binary outcome, and the linear model gives the most straightforward coefficient interpretation (Angrist and Pischke 2009). The first specification regresses the species-reporting indicator on an indicator for whether a sample comes from a contract area:

$$\text{Species Reported}_{ict} = \text{Intercept} + \beta \cdot \text{ContractArea}_i + \varepsilon_{ict} \quad (10)$$

The coefficient of interest, β , captures the difference in the mean probability of species-level identification between contract-area samples and non-contract-area samples (APEIs and outside areas), without accounting for confounders. Each observation i is a biological specimen record from contractor c in year t , and ε_{ict} is the error term. A negative β indicates that specimens from contract areas are less likely to carry a species-level identification than specimens from non-contract areas.

The base specification omits potential confounders, so the estimated β could reflect factors other than strategic behavior. First, differences in taxonomic resolution could be driven by differences in scientific methodology rather than incentives. To absorb this, I add fixed effects for the mining contractor and for the

year. Contractor fixed effects absorb average differences in reporting rates across the 12 contractors in the data—each a country-backed entity with an ISA exploration contract for one or more tracts (e.g., Nauru Ocean Resources Inc.). This matters because contractors that generally report more taxonomic detail might also be the ones submitting more non-contract-area samples, which would generate a spurious relationship in the raw data. Year fixed effects absorb average differences in reporting rates by year, accounting for improvements in scientific methods and taxonomic knowledge over time. The corresponding regression equations are:

$$\text{Species Reported}_{ict} = \beta \cdot \text{ContractArea}_i + \text{Contractor FE}_c + \varepsilon_{ict} \quad (11)$$

$$\text{Species Reported}_{ict} = \beta \cdot \text{ContractArea}_i + \text{Contractor FE}_c + \text{Year FE}_t + \varepsilon_{ict} \quad (12)$$

A second possible confounder is ecological. Non-contract areas and prospectively-mined tracts could differ in conditions that affect species-identification rates, and the threat is most acute if APEIs were placed where mining is unattractive *because* the biology there is unusual—producing a comparison group that is systematically not like the contract tracts. The ISA’s siting principles are designed to do the opposite. APEIs are large rectangular blocks (most are 160{,}000~km²) sited under an explicit “representativity” criterion: they are intended to span the major environmental gradients of the CCZ so that whatever fauna are eventually documented there resemble what would be displaced by mining elsewhere (Wedding et al. 2013; Smith et al. 2021). They were carved out of the CCZ grid rather than chosen to surround unusual habitats. APEIs are, by design, the closest analogue to contract tracts that the regulatory architecture allows. The available evidence is broadly consistent with this—\/-\/-@washburnEnvironmentalHeterogeneityThroughout2021 find that the APEIs are environmentally representative of the contract areas on many dimensions, though not all: most notably, the contract areas were selected for their mineral value and so hold denser nodules than the APEIs. To absorb that residual heterogeneity, I add four environmental controls that capture the major axes of variation across the CCZ: depth (which influences pressure, habitat type, and benthic community composition), sea surface temperature (the east-west and north-south productivity gradients), surface chlorophyll concentration (a direct measure of primary productivity and thus the organic carbon flux that sustains abyssal communities), and dissolved bottom oxygen (which constrains benthic metabolic capacity). All environmental data are from Bio-ORACLE rasters matched to individual sample locations. Variance inflation factors for the environmental controls are all below conventional thresholds for concern. Nodule abundance, another potentially relevant variable, is omitted because spatially resolved nodule data

are not publicly available at the sample level; the contractor fixed effects partially absorb nodule-related variation, since prospectively-mined tracts were allocated in part on the basis of nodule density. The fourth regression equation is then:

$$\text{Species Reported}_{ict} = \beta \cdot \text{ContractArea}_i + \text{Environmental Controls}_i + \text{Contractor FE}_c + \text{Year FE}_t + \varepsilon_{ict} \quad (13)$$

where environmental controls include sea surface temperature, surface chlorophyll, dissolved bottom oxygen, and depth matched to each sample’s location.

A remaining concern is that differences in species reporting could reflect differences in the *composition* of organisms sampled rather than strategic behavior. If contract areas happen to contain taxa that are inherently harder to identify to the species level, the gap could arise even without deflation. Importantly, taxonomy is itself endogenous to the exploration process: contractors who deflate will report coarser taxonomic classifications, so conditioning on taxonomy absorbs variation that may itself be generated by the deflation mechanism. Nonetheless, as a robustness exercise I present results from specifications that each condition on a single taxonomy fixed effect (phylum, class, order, family, or genus) on top of the full set of controls. Each specification restricts the sample to observations with a valid identification at the relevant taxonomic level. This is a conservative test: if the gap survives even after conditioning on endogenous taxonomy at the finest level (genus) it cannot be attributed to compositional differences in the organisms sampled at any taxonomic level, but it can also understate gaps in species reporting that are due to identifications that stopped at even higher taxonomic levels.

Two further robustness exercises exploit a structural feature of the dataset uncovered by overlaying official ISA contract polygons on each observation’s coordinates: a small but consistent share of biological samples are collected by one contractor inside *another* contractor’s polygon, typically during joint expeditions or transit segments. These “guest” samples are processed by the same scientific institutions as own-tract samples, but carry no permit-related incentive for the sampler. The structure enables (i) a within-sampler design comparing each contractor’s own-tract reporting to its guest-tract reporting (holding scientific capacity fixed) and (ii) a within-polygon design comparing owner-collected to guest-collected samples inside the same physical tract (holding ecology fixed). Together the two designs sandwich the headline effect from above and below using identifying variation that the contractor-and-environmental controls in the main regressions cannot use.

The four cross-zone specifications are reported as a ladder rather than collapsed into a single preferred

model because each column trades off two desirable properties. The baseline column uses the full sample of contractor submissions and treats every observation symmetrically; in particular, it includes the substantial number of contractors who only ever submit data from inside their own contract tract, whose comparison group is mechanically other firms’ guest visits to their tract and APEI material. These contractors contribute most of the variation in the dependent variable, but contribute none of the variation in the treatment indicator within their own observations. Successively adding contractor, year, and environmental fixed effects forces the contract-area coefficient to be identified off increasingly demanding contrasts: within the same firm in the same year facing the same temperature, depth, and productivity. The cost is that the sample restricts and the within-firm comparison group shrinks. Reporting all four specifications lets the reader weigh that tradeoff directly; the headline effect is the within-firm-within-year-within-environment estimate (column 4), but the columns to its left document that the cross-firm comparison points in the same direction on the larger and more representative sample.

Across all specifications, if contract-area samples are reported with lower taxonomic resolution after accounting for confounders, β should be negative and statistically significant.

5 Results

5.1 Exploration Intensity across Zones

The model’s first prediction, that contractors conduct more physical sampling on prospectively-mined tracts than on non-contract areas, can be evaluated directly from the DeepData records before any regression analysis. Figure 2 reports both the raw count of biological specimens by zone and the per-square-kilometre sampling intensity, using polygon areas computed from the ISA’s official PMN and APEI shapefiles. Both views confirm Prediction 1: contract areas account for the overwhelming majority of specimens in absolute terms *and* receive sampling at a per-area rate that is hundreds of times higher than APEIs.

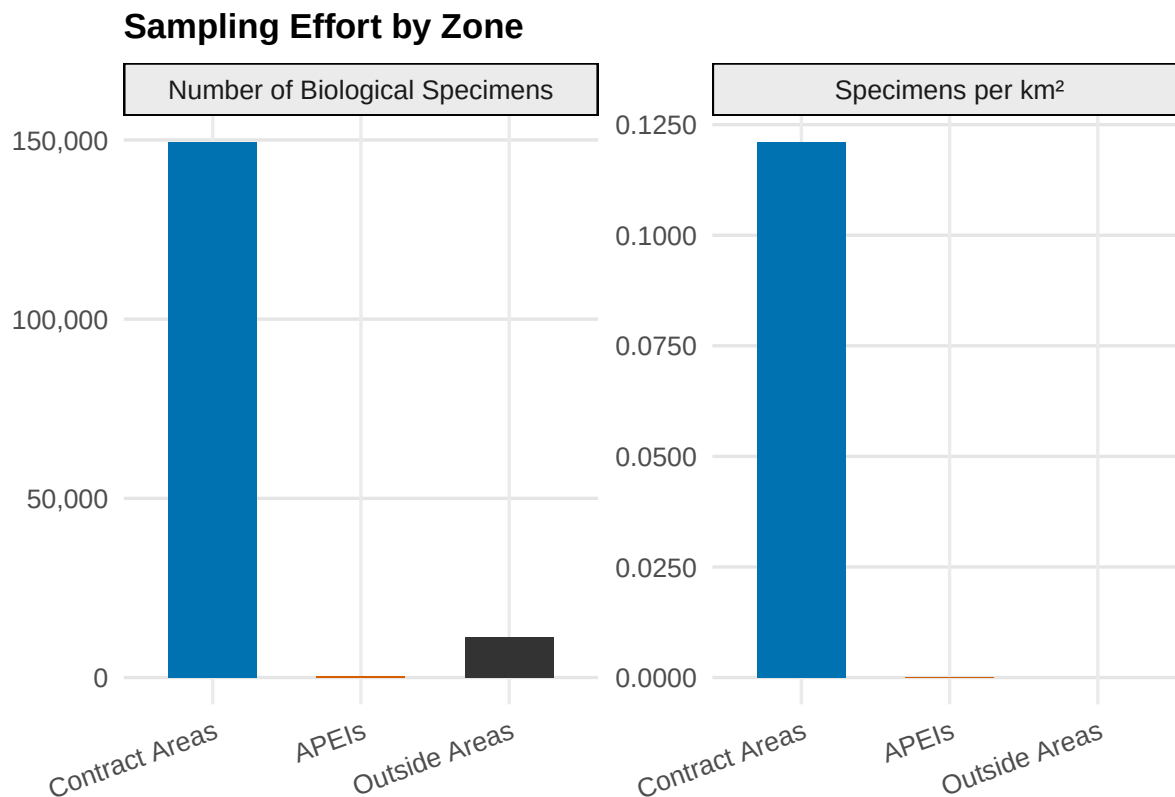


Figure 2: Sampling effort by zone, raw counts (left) and per-square-kilometre intensity (right). Sample counts are taken from the analysis sample; areas are computed from the ISA’s official PMN and APEI shapefiles projected to a cylindrical equal-area projection centred at 10°N. APEIs and contract areas have comparable aggregate footprint in the CCZ (~ 2.0 vs. ~ 1.2 million km²), so the per-area panel is a like-for-like intensity comparison. The ‘Outside’ bar in the right panel is omitted because there is no well-defined denominator for samples falling outside both contract polygons and APEIs.

Consistent with Prediction 1, the overwhelming majority of biological samples in DeepData come from contract areas rather than non-contract areas, and the gap survives normalizing by the underlying footprint of each zone. Contract-area sampling intensity is 0.121 specimens per km² against 0.0002 per km² in APEIs—a ratio of roughly 663-to-one, even though contract areas and APEIs are of comparable aggregate size in the CCZ (1,234,080 km² of PMN polygons against 1,962,328 km² of APEIs). This is what the model predicts: the compliance value that sampling buys is worth paying for only where a permit hangs on it. The asymmetry shows up beyond physical sampling. The administrative metadata fields that document the post-collection workflow (e.g. whether a taxonomist was involved, the identification method, the identification

date, etc) are filled in for roughly 70% of contract-area records but for fewer than 3% of non-contract-area records. Contractors invest substantially more effort in both field collection and compliance documentation on samples that bear on their own permit. Appendix Table 12 reports the details. The critical question is whether this greater exploration effort translates into correspondingly greater taxonomic detail. Under a pure resource-constraints explanation, the zone with more samples should also have better species coverage. The model’s second prediction is that the opposite holds.

5.2 Taxonomic Reporting by Zone

The raw data confirm Prediction 2. Figure 3 shows that most taxonomic levels are more likely to be reported in non-contract areas. For example, 44.3% of observations from non-contract areas report the species of the sample, against just 16.4% in contract areas. The raw difference could reflect many things—I show below that it persists after the obvious controls.

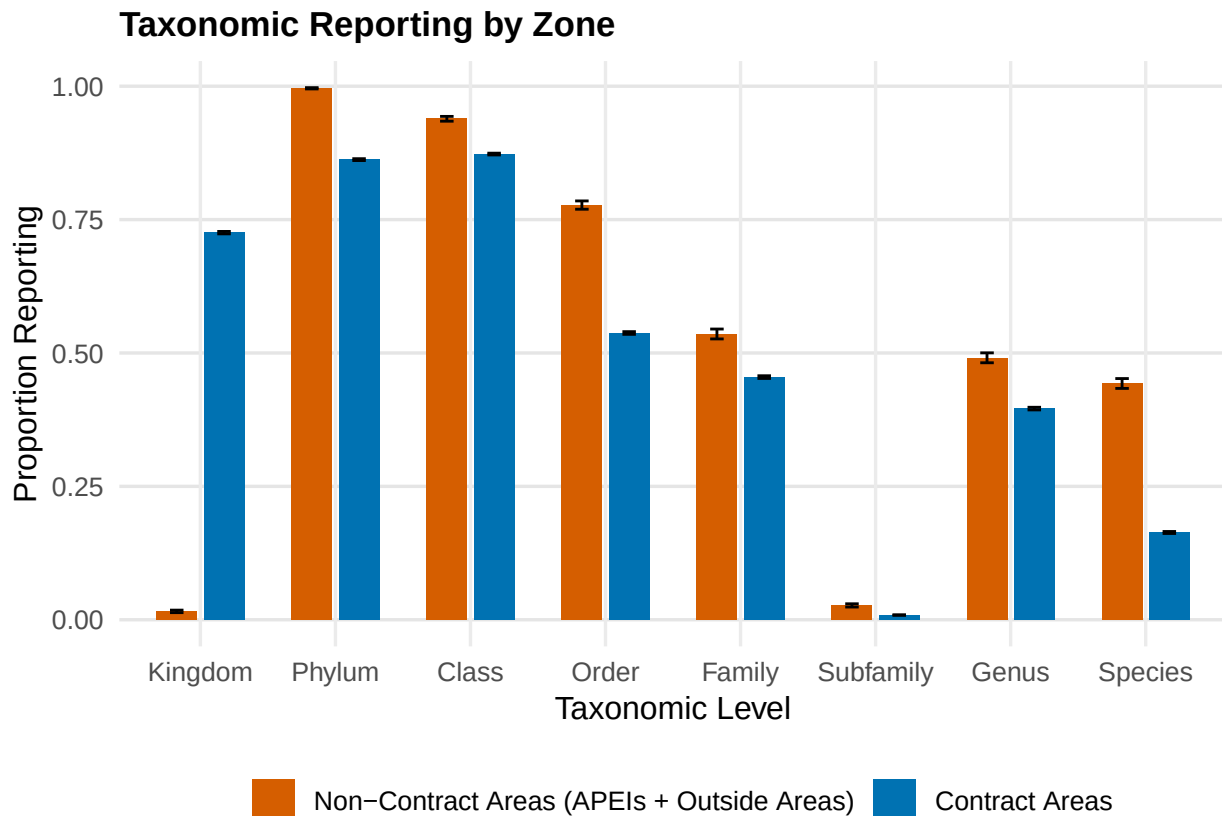


Figure 3: Proportion of specimen records with a non-missing value at each taxonomic level. Each level is evaluated independently from the others: a record counts as reported at a given level if and only if its corresponding taxonomic field (Kingdom, Phylum, . . . , Species) contains a non-missing value, regardless of whether coarser or finer levels are also populated. The gap widens at finer taxonomic levels, consistent with strategic deflation at the margins where species discovery poses the greatest regulatory risk. 95% confidence intervals shown.

5.3 Regression Results

The regressions provide a formal test of Prediction 2. Figure 4 plots the estimated $\hat{\beta}$ —the precision gap $\tau_M - \tau_A$ from the model—and 95% confidence intervals for four specifications that incrementally add exogenous controls. The coefficient on contract-area status is negative and statistically significant in four of the four designs, implying anywhere from 28 to 6 percentage points lower probability of species reporting in contract areas relative to non-contract areas; the preferred specification (column 4, which adds environmental controls within firm and year) puts the gap at -0.060 . Consistent with the model, contractor fixed effects—which absorb variation in scientific capacity across firms—reduce but do not eliminate the gap (Appendix Figures

11 and 12 summarize sampling volume and species reporting by contractor). Adding year fixed effects shrinks the coefficient further ($p = 0.000$), but the gap remains negative and significant at conventional thresholds; environmental controls sharpen precision further. Full regression results are reported in Table 6 in the Appendix. Appendix Tables 9 and 10 report a complementary exercise that replaces the binary species-reporting outcome with a graduated *stopping level* (0–7, the deepest taxonomic level each record reaches). Conditional on the same controls, contract-area records stop on average about two-thirds of a taxonomic level shallower than non-contract-area records, implying the gap is not concentrated at the species threshold but rather it is a graduated tendency to stop earlier in the taxonomic hierarchy.

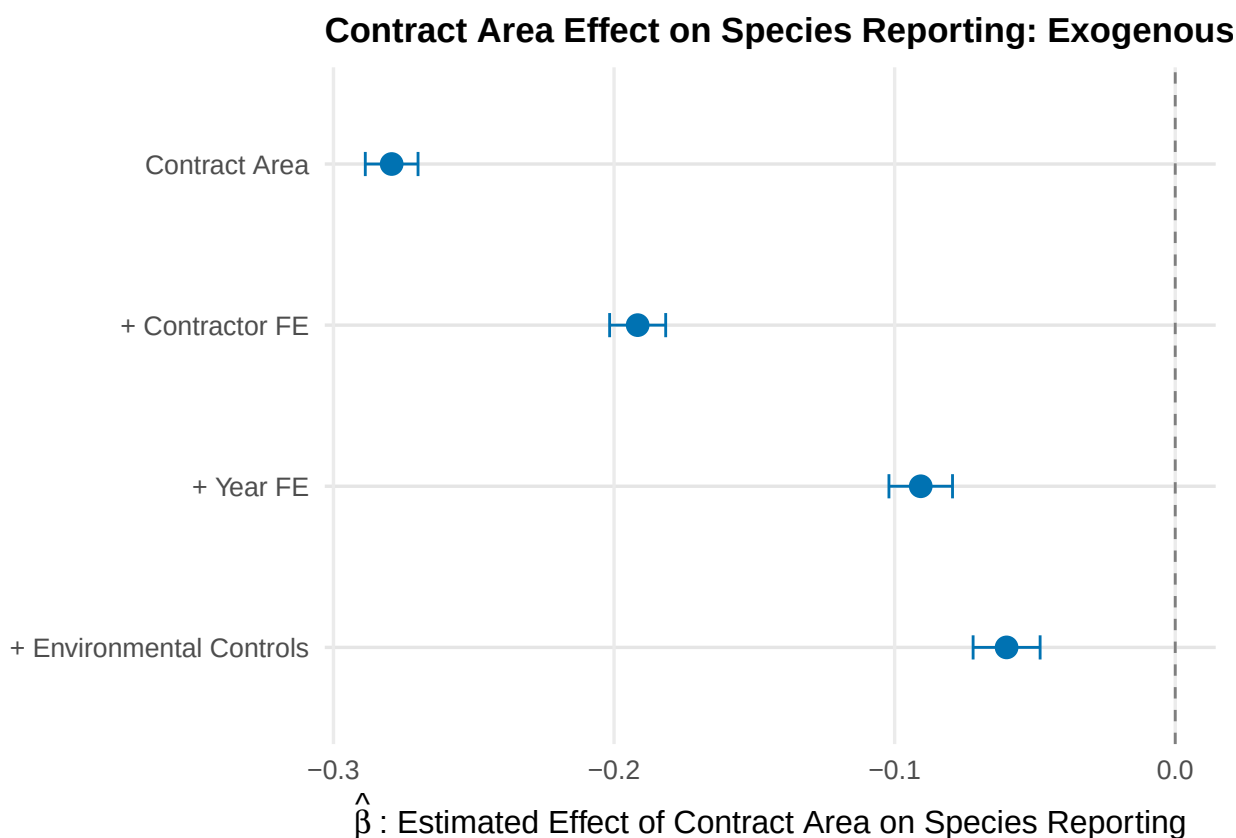


Figure 4: Estimated contract area coefficient ($\hat{\beta}$) from four LPM specifications, each adding incrementally more confounding variable controls. Each point represents the estimated difference in the probability of species-level identification for contract area samples relative to non-contract area samples (APEIs and outside areas). Negative coefficients indicate that species are less likely to be reported from a contract area than from a non-contract area. 95% confidence intervals computed with robust standard errors.

5.4 Robustness to Taxonomic Composition

A natural concern is that the gap reflects compositional differences in the organisms sampled across zones rather than strategic behavior. If contract areas happen to contain taxa that are inherently harder to identify, the reporting gap could arise without deflation. To address this, Figure 5 presents a taxonomy robustness exercise that separately conditions on each taxonomic level (from phylum through genus) on top of the full set of exogenous controls (contractor, year, and environmental variables). Taxonomy is itself endogenous to the exploration process—contractors who deflate report coarser taxonomic classifications—so conditioning on taxonomy is a conservative test that absorbs variation generated by the deflation mechanism itself. Even so, the contract-area coefficient remains negative and statistically significant when I compare organisms within the same genus, ruling out compositional differences at every taxonomic level as an explanation for the gap (Table 7).

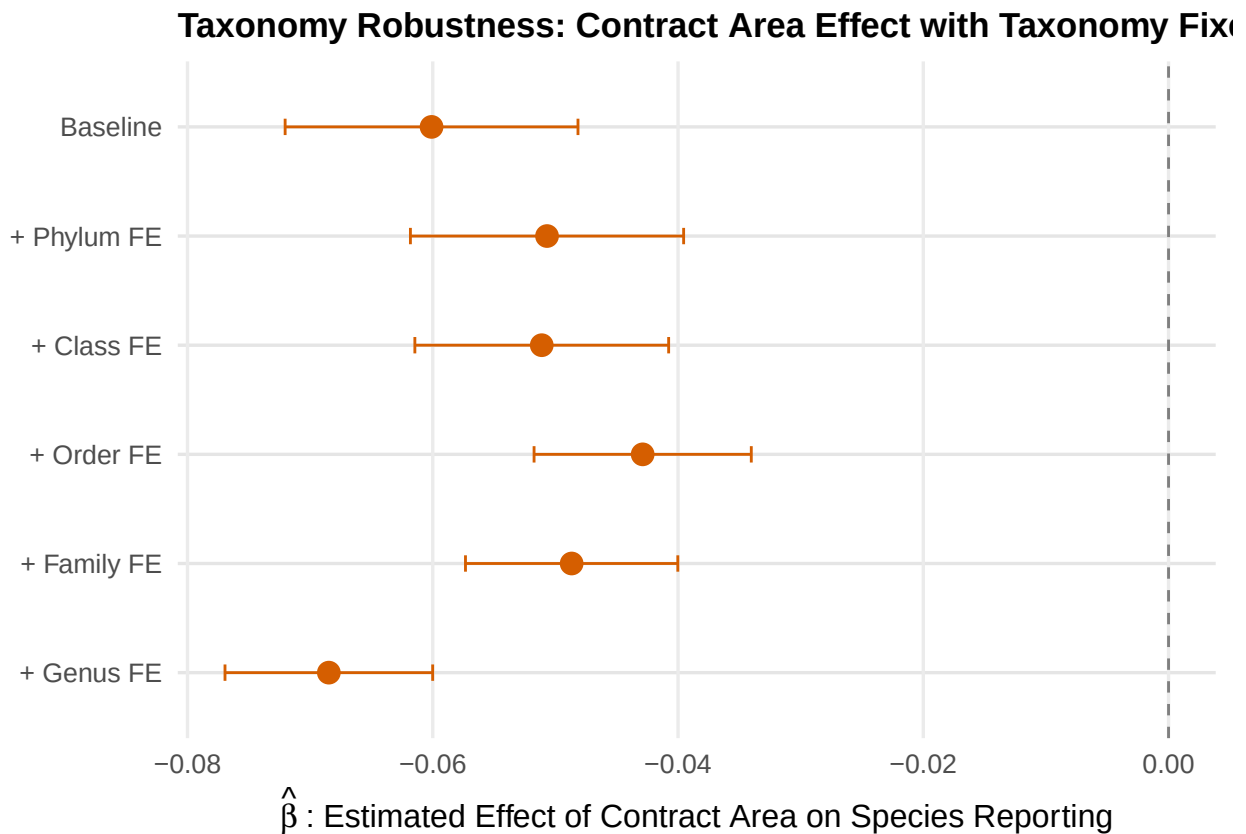


Figure 5: All specifications include contractor, year, and environmental controls. Each specification restricts the sample to observations with a valid identification at the relevant taxonomic level. 95% confidence intervals shown.

The persistence of $\hat{\beta} < 0$ after controlling for contractor identity, year, environmental conditions, and taxonomy indicates that the gap in species-level reporting between contract areas and non-contract areas is robust to a wide range of potential confounders. I take up the interpretation of this pattern, including alternative explanations, in the following sections.

A caveat: species-level identification in the deep sea is genuinely difficult—an estimated 88–92% of CCZ metazoans are undescribed (Rabone, Wiethase, et al. 2023) and even expert teams identify fewer than 20% of observed species (Kennedy et al. 2019)—and I do not claim that every coarse identification reflects strategic behavior. But that difficulty is common to both zones, so it cannot explain the systematic *between-zone* gap. I take up inherent difficulty, and the other non-strategic explanations, in the Discussion.

5.5 Within-Sampler and Within-Polygon Robustness

The contractor and taxonomy fixed effects above absorb cross-firm differences in scientific capacity and cross-taxon differences in identifiability, but they do not directly hold the *physical location* of each sample fixed. Geological and ecological conditions vary in space, and if contractor tracts are systematically located in regions where species are harder to identify for non-strategic reasons, the regression results above could be confounded. I address this by exploiting a structural feature of the dataset that emerges when I overlay the official ISA contract polygons onto each observation’s coordinates: a small but consistent share of biological samples are collected by one contractor inside *another* contractor’s polygon, typically during joint expeditions or transit segments. These “guest” samples are processed by the same scientific institution, in the same laboratory pipeline, and often during the same cruise as the sampler’s own-tract samples—but they carry no permit-related incentive for the sampling contractor, since the sampler does not hold mining rights at the location where the specimen was collected. Appendix Table 8 documents the full source decomposition.

This cross-contractor sampling enables two distinct robustness designs, each holding fixed a different confounder:

1. *Same contractor, different areas*: a within-sampler comparison that asks whether a given contractor reports samples differently when collecting in their own tract versus when collecting in another contractor’s tract. Sampler fixed effects absorb cross-firm differences in scientific capacity, so the surviving variation isolates the regulatory regime change.
2. *Same area, different contractor*: a within-polygon comparison that asks whether, inside a single physical tract, samples collected by the polygon owner differ from samples collected by guest contractors. Polygon fixed effects absorb local ecology, so the surviving variation isolates the sampler-by-regime contrast at

that location.

The two designs are complementary. The first holds reporting infrastructure fixed at the cost of leaving ecology to vary; the second holds ecology fixed at the cost of leaving reporting infrastructure to vary. Both deliver estimates that support the headline finding, and I report them in Tables 1 and 2 below. Both designs use as the sample all observations inside some PMN polygon, across all 12 sampling contractors. Samplers without Owner_{*i*} variation contribute nothing to the within-sampler $\hat{\beta}$, and polygons without Owner_{*i*} variation contribute nothing to the within-polygon $\hat{\beta}$, but those observations remain in the regression to inform the year and environmental coefficients. The estimating equations share a common structure:

$$\text{Species Reported}_{ipct} = \beta \cdot \text{Owner}_{ipct} + \mu_p + \gamma_c + \delta_t + X'_{ipct}\eta + \varepsilon_{ipct} \quad (14)$$

where Owner_{*i*} = 1 if observation *i* was collected by the polygon owner and 0 if by a guest. Different combinations of fixed effects identify β from different variation: with sampler fixed effects γ_c but not polygon fixed effects μ_p , β captures the within-sampler difference in species reporting between own-tract and guest-tract samples; with polygon fixed effects but not sampler fixed effects, β captures the within-polygon difference between owner-collected and guest-collected samples; including both is uninformative because polygon and ownership are nearly collinear (each polygon has exactly one owner).

Within-sampler estimates are reported in Table 1. Column (1) reports the raw owner-vs-guest gap, $\hat{\beta} = -0.292$ —here $\hat{\beta}$ is the coefficient on the Owner indicator, which plays the role of ContractArea in these designs, since an owner-collected sample is the permit-relevant one. Owner-collected specimens are roughly 29 percentage points less likely to carry a species-level identification than guest-collected specimens. Adding sampler fixed effects shrinks the estimate to -0.231 , year fixed effects further to -0.104 , and environmental controls leave it at -0.104 (SE 0.006). The within-sampler estimate is somewhat larger in magnitude than the main paper’s preferred specification ($\hat{\beta} = -0.060$ in Figure 4 and column (4) of Table 6), but recovered from a substantially more demanding identification—only within-sampler, within-year variation between own-tract and guest-tract samples is used.

Within-polygon estimates are reported in Table 2. Holding the physical tract fixed with polygon fixed effects alone yields $\hat{\beta} = -0.203$, roughly twice the within-sampler estimate; adding year fixed effects roughly doubles this further to -0.400 , and environmental controls leave it at -0.398 (SE 0.007). The fully-controlled within-polygon estimate is roughly four times the within-sampler estimate. The expansion when I add year fixed effects reflects time variation in cross-sampler reporting practices—guest-tract samples are concentrated in particular years and are collected by contractors with above-average reporting rates, so within-polygon,

Table 1: Within-sampler design (same contractor, different areas): species reporting on Owner indicator, holding scientific capacity fixed via sampling-contractor fixed effects. Sample: all observations inside some PMN polygon, across all 12 sampling contractors. Samplers without Owner_i variation drop from $\hat{\beta}$ identification. Robust (HC1) standard errors in parentheses.

	Owner	+ Sampler FE	+ Year FE	+ Env. Controls
Intercept	0.467*** (0.005)	0.284*** (0.006)	1.495*** (0.009)	-47.519*** (2.371)
Owner sampler	-0.292*** (0.005)	-0.231*** (0.006)	-0.104*** (0.006)	-0.104*** (0.006)
Depth (m)				0.000*** (0.000)
Sampler FE	No	Yes	Yes	Yes
Year FE	No	No	Yes	Yes
Environmental Controls	No	No	No	Yes
Sea Surface Temp.				0.628*** (0.030)
Surface Chlorophyll				-4.758*** (0.512)
Dissolved Oxygen				6.855*** (0.341)
Num.Obs.	149 300	149 300	149 300	149 300
R2	0.031	0.110	0.211	0.217

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

within-year contrasts magnify the gap between owner and guest. The larger magnitude also reflects the fact that the within-polygon comparison picks up cross-sampler differences in scientific capacity at the same location alongside the regulatory regime change, so the fully-controlled within-polygon estimate is best read as an upper bound on the deflation effect under the assumption that residual within-polygon contractor-level heterogeneity reflects the regulatory regime rather than independent capacity differences. The two designs are best read as separate, mutually-reinforcing exercises. That is, the headline finding survives whichever confounder one is most worried about: holding scientific capacity fixed gives a -0.104 effect, holding ecology fixed gives a -0.398 effect. The within-sampler design plausibly understates the true deflation effect (guest-collected samples may receive less intensive post-processing than own-tract samples regardless of regulatory regime); the within-polygon design plausibly overstates it (cross-firm differences in scientific capacity contribute alongside the regulatory mechanism).

The identifying variation behind the within-sampler estimate is visible in Figure 6, which plots, for each of the five contractors who appear as both polygon owners and as guests, the species-reporting rate of their own-tract samples against the species-reporting rate of their guest-tract samples. The pattern is consistent across firms with substantial guest-tract sample sizes: BGR reports 6% of its own-tract samples to species against 27% of its guest-tract samples ($n=2,378$); UKSRL reports 30% against 54% ($n=6,553$). The other

Table 2: Within-polygon design (same area, different contractor): species reporting on Owner indicator, holding ecology fixed via polygon fixed effects. Sample: all observations inside some PMN polygon. Polygons without Owner_i variation drop from $\hat{\beta}$ identification. Robust (HC1) standard errors in parentheses.

	Owner	+ Polygon FE	+ Year FE	+ Env. Controls
Intercept	0.467*** (0.005)	0.262*** (0.008)	1.695*** (0.012)	-102.612*** (1.943)
Owner sampler	-0.292*** (0.005)	-0.203*** (0.008)	-0.400*** (0.007)	-0.398*** (0.007)
Depth (m)				0.000*** (0.000)
Polygon FE	No	Yes	Yes	Yes
Year FE	No	No	Yes	Yes
Environmental Controls	No	No	No	Yes
Sea Surface Temp.				0.949*** (0.025)
Surface Chlorophyll				-2.836*** (0.498)
Dissolved Oxygen				16.692*** (0.321)
Num.Obs.	149 300	149 300	149 300	149 300
R2	0.031	0.094	0.209	0.219

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

three contractors with both roles (GSR, IFREMER, Korea) have very thin guest-tract samples (n=21 to 52) and should not be over-interpreted; for example, Korea's apparent reversal is identified off just 52 guest observations.

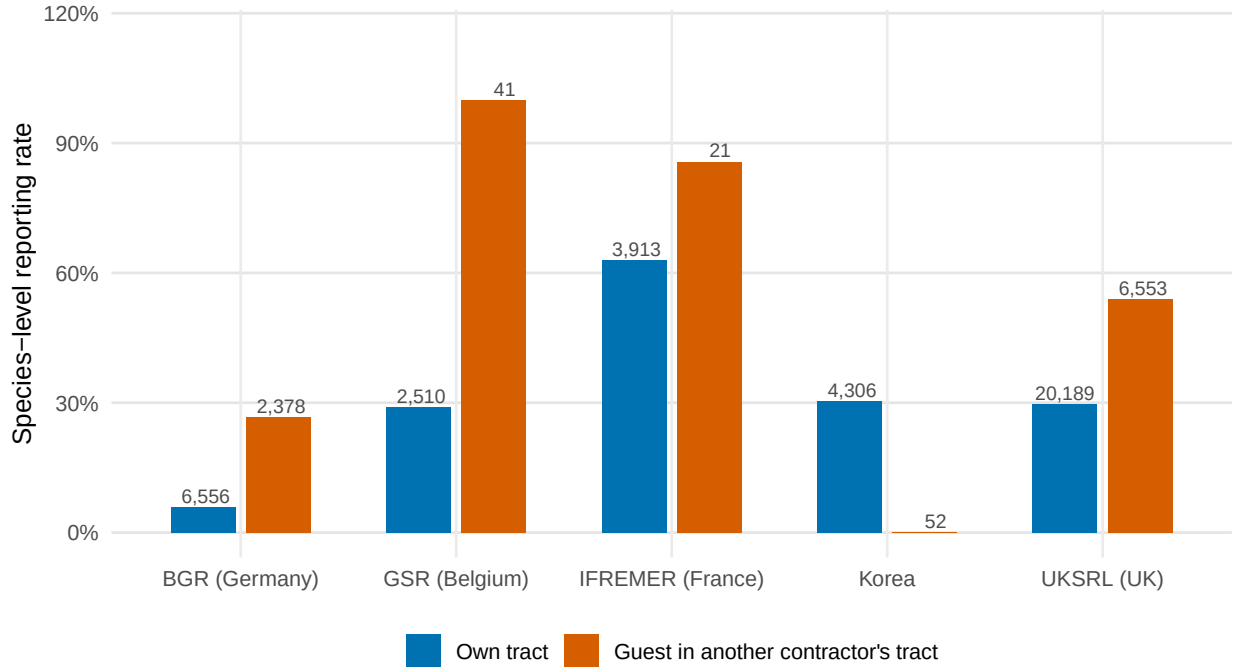


Figure 6: Species-level reporting rate, by sampling contractor and own-vs-guest status. Restricted to the five contractors who appear as both polygon owners and as guests in another contractor’s polygon. Numbers above each bar are the underlying sample sizes; bars with very small n (Korea, GSR-guest, IFREMER-guest) should be interpreted with caution. The within-sampler estimate in Table `reftab:reg-table-within-sampler` averages these contrasts.

5.6 Mechanisms: Cruise-Level Selection, Identifier Assignment, and the Identifying Institution

The headline regression and the within-sampler and within-polygon designs document the existence of a deflation gap and rule out several confounders. They do not yet speak to the *channel* through which the gap operates: do contractors systematically organize their sampling and identification work differently for permit-relevant tracts? Two structural features of the data let me distinguish two candidate channels.

The first is the cruise leg. Biological samples in DeepData are organized by expedition, each corresponding to a single research cruise with a particular vessel, scientific team, and operational window. The dataset spans 44 cruise legs in the analysis sample. If the deflation gap reflects between-expedition selection—contractors deploying different teams or methods on different cruises—then absorbing cruise-leg fixed effects should attenuate the gap.

The second is the named individual taxonomist who produced each identification. 84% of records in the analysis sample report an identifier name, covering 84 distinct taxonomists. Adding identifier fixed effects then asks whether *the same person* produces different reporting outcomes for contract-area and non-contract-area samples, holding individual expertise and methodology fixed by construction.

Table 3 reports the results.

Table 3: Mechanism analyses. Baseline specification (Spec 4 of Table \@ref(tab:reg-table)) adds cruise-leg fixed effects in columns (1)–(2) and identifier-name fixed effects in columns (3)–(4). Column (2) excludes NORI, which dominates the data and whose cruises have minimal cross-zone variation. Column (4) restricts to identifiers who appear with both contract and non-contract observations. All specifications include contractor, year, and environmental controls. Robust (HC1) standard errors in parentheses.

	+ Cruise-leg FE	+ Cruise FE (no NORI)	+ Identifier FE	+ Identifier FE (balanced)
Contract Area	−0.028*** (0.005)	−0.020*** (0.005)	−0.001 (0.006)	0.002 (0.005)
Num.Obs.	150 104	54 907	135 381	21 686
Contractor FE	Yes	Yes	Yes	Yes
NA	NA	NA	NA	NA
Year FE	Yes	Yes	Yes	Yes
Environmental Controls	Yes	Yes	Yes	Yes
Cruise-leg FE	Yes	Yes	No	No
Identifier FE	No	No	Yes	Yes
R2	0.431	0.594	0.560	0.555

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Adding cruise-leg fixed effects to the main specification shrinks the contract-area coefficient from -0.060 (column 4 of Table 6) to -0.028 (SE 0.005). About half of the headline effect is absorbed by between-cruise variation: different expeditions sample different zones with different scientific teams, vessels, gear, and processing pipelines. Whether this between-cruise component is itself a confound to be controlled away or a *first stage* of the deflation mechanism is a matter of interpretation—contractors choose which cruises go where, with what teams and equipment, and those choices are themselves part of how the firm allocates identification effort across zones. The cruise-leg specification therefore does not test whether deflation is real; it bounds the share of the headline gap that operates *conditional on* the contractor’s cruise-level decisions. The within-cruise estimate of -0.028 is best read as a lower bound on the deflation that operates within a single expedition—that is, the part of the gap that cannot be attributed to cruise-level selection. Excluding NORI (whose data dominate the sample and whose cruises have minimal cross-zone variation) yields a comparable residual estimate of -0.020 , and both within-cruise estimates remain statistically significant at $p < 0.001$.

Adding identifier-name fixed effects collapses the deflation gap further. With identifier fixed effects in place, the contract-area coefficient is -0.001 (SE 0.006)—statistically indistinguishable from zero. The same is

true when I restrict the sample to identifiers who produced records in both contract and non-contract areas (column 4, $\hat{\beta} = 0.002$, SE 0.005). That is, the same individual taxonomist produces statistically identical species-reporting rates across contract and non-contract samples.¹² Appendix Table 11 reports a selection-bias check that re-runs the headline specification on the identifier-FE sub-samples *without* identifier fixed effects, distinguishing the share of the attenuation that comes from FE absorption from the share that comes from operating on a non-representative sub-sample.

If the gap lives in how samples get assigned, the laboratories doing the identifications are the natural place to look next. About 85% of records name an identification institution, and these cluster in a few labs with sharp specialties—foraminifera at one, nematodes at another, fishes and large crustaceans at a third, and so on. That specialization makes raw comparisons across institutions misleading, because taxa differ enormously in how often they can be named at all (nine in ten chordate records reach species level, against fewer than one in a hundred for corals or bryozoans). For example, a lab that works mostly on fish will post a high species rate whatever its effort because fish species are easier to identify for a variety of reasons. To isolate reporting behavior, the comparison has to hold the taxon fixed within the institution.

Table 4: The identifying institution and the deflation gap. Column (1) reproduces the baseline specification (Spec 4 of Table \@ref(tab:reg-table)); columns (2)–(5) restrict to records that name an identification institution and progressively add institution and taxonomy fixed effects. Comparing specimens of the same taxon identified by the same institution (columns 4–5) isolates within-laboratory reporting. Robust (HC1) standard errors in parentheses.

	Baseline	Inst. named	+ Institution FE	+ Institution + Phylum FE	+ Institution + Genus FE
Contract Area	−0.060*** (0.006)	−0.018** (0.007)	−0.021*** (0.007)	−0.053*** (0.006)	−0.040*** (0.003)
Contractor FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Environmental Controls	Yes	Yes	Yes	Yes	Yes
Institution FE	No	No	Yes	Yes	Yes
Phylum FE	No	No	No	Yes	No
Genus FE	No	No	No	No	Yes
Num.Obs.	150 104	127 271	127 271	125 509	62 883
R2	0.211	0.289	0.536	0.565	0.849

* p < 0.1, ** p < 0.05, *** p < 0.01

Institution fixed effects on their own barely touch the gap (Table 4): it moves from −0.018 to −0.021 once I keep only records that name a lab, so which laboratory handles a sample is not what drives it. The picture changes once taxon is held fixed too. Comparing specimens of the same phylum or the same genus identified by the same institution, the gap is −0.053 and −0.040 respectively, both strongly significant, and this within-laboratory, within-taxon estimate survives dropping any single institution. Like for like, same

¹²The identifier-name field is populated for 90.2% of the analysis sample, and the balanced sub-sample restricts to identifiers who appear in both zones (21,686 observations). Sample selection between the full analysis sample and the identifier-present sub-sample is therefore a live concern for interpreting the FE attenuation.

lab and same taxon, contract-area samples reach species level some 4 to 5 percentage points less often. The gap is not an accident of the database’s unattributed records: it shows up among the fully-documented ones, and the records that omit an institution carry much the same gap (-0.024), so it is not hiding in the metadata-poor entries. And it nests with the identifier result—the gap survives controls for both the institution and the taxon, but is absorbed once the comparison narrows to a single identifier.

Read together, these analyses locate the mechanism. The within-cruise component of the deflation gap is essentially explained by *which* taxonomist gets assigned to which sample. Contractors do not pressure individual taxonomists into producing different work for permit-relevant samples; they simply route less-precise identification capacity to those samples instead. That is, the same outcome—coarser data from permit-relevant zones—is produced not by individual misreporting but by systematic resource allocation across the contractor’s identification pipeline. The distinction matters for policy, as the assignment channel can be addressed by instruments that decouple identifier assignment from contractor discretion (independent taxonomic review, sealed reporting, certified-expert assignment), as I discuss further in Section 7.

6 Discussion

The pattern I document fits the principal-agent problem the model describes: the ISA delegates biodiversity assessment to firms whose returns depend on subsequently being allowed to mine, and effort along the precision dimension is hard to monitor. The model makes predictions on two margins, and the data bear out both. Contractors sample the tracts they intend to mine far more intensively than the non-contract areas, yet report those denser collections with systematically coarser taxonomy. More effort paired with less precision in the same place is the joint signature the model singles out, and it resists any account in which reporting quality is capped by resources or capacity—there, effort and precision would rise together, not diverge. Two features of the empirical evidence are central to the strategic interpretation. First, the gap survives the most demanding controls the data allow—contractor identity, year, environmental conditions, taxonomy down to genus, and the scientific capacity of both the sampling firm (in the within-sampler exercise) and the identifying laboratory (in the within-institution analysis). Second, both spatial robustness designs point in the same direction. When the *same* contractor samples in another firm’s tract, it reports more precisely than in its own tract; when the *same* tract receives samples from both its owner and a guest contractor, the owner’s samples are coarser. No non-strategic mechanism predicts both patterns. My empirical design documents an association rather than directly observing intent, but the alternatives I consider below cannot account for the joint within-sampler-and-within-polygon evidence.

6.1 Alternative Explanations

Two of the most obvious confounds are already dispatched by direct evidence above—compositional differences in the organisms sampled (the taxonomy-fixed-effects results, Figure 5) and spatial confounding (the within-sampler and within-polygon designs)—so I do not revisit them here. The alternatives that remain cannot be settled by any single specification, and I take them up in turn.

The most prominent objection is that the reporting gap reflects the inherent difficulty of deep-sea taxonomy rather than strategic behavior. Deep-sea identification is genuinely hard—less than 0.001% of the seafloor has been visually surveyed (Bell et al. 2025), expertise is scarce (Frutos et al. 2022), and an estimated 88–92% of CCZ species remain formally undescribed (Rabone, Wiethase, et al. 2023). Those constraints almost certainly explain why species-level reporting is low in *absolute* terms across the dataset.¹³ What they cannot explain is why reporting rates differ *systematically* between zones, given that every sample in the analysis was collected from the abyssal CCZ and faces the same taxonomic knowledge gaps. If inherent difficulty were the sole driver, it would depress species reporting symmetrically across contract areas and non-contract areas. The gap between zones is precisely the component that science alone cannot explain.

A second alternative is that the gap reflects differences in *survey objectives* between zones. Non-contract-area surveys may be designed with biodiversity documentation as a primary goal and receive greater post-cruise taxonomic follow-up, while surveys of contract areas may prioritize environmental baseline characterization. If non-contract-area samples are more likely to be sent to specialized taxonomic laboratories or to receive extended analysis by academic collaborators, species-level identification rates could be higher even without any difference in field effort. Contractor fixed effects partially address this by absorbing average differences in each contractor’s scientific methodology and institutional partnerships, but they cannot absorb *within-contractor* differences in how samples from different zones are processed after collection. If a contractor systematically invests more in the taxonomic analysis of its non-contract-area samples than its contract-area samples, that pattern would generate the observed gap. I cannot directly test for differential post-collection processing, but the direction of this explanation is itself consistent with the incentive story: a contractor that allocates more taxonomic effort to non-contract-area samples is responding to the same asymmetric incentives the model

¹³Independent academic surveys of the CCZ provide a useful calibration of what species-level taxonomy looks like when commercial considerations are absent. Focused taxonomic campaigns—such as the Sustainable Seabed Knowledge Initiative workshop in Jazdzewska and Horton (2026), which described 24 new amphipod species (across ten families, one new family, and one new superfamily) from CCZ material—routinely reach the species level, often as new-to-science but still with a binomial name and a formal description that would register as species-level reporting in DeepData. Earlier CCZ work, including Bonifácio et al. (2021) on Polynoidae and Stewart et al. (2026) on broader megafauna, follows the same pattern: when an expedition is designed around taxonomic documentation rather than permit compliance, the working norm is to push specimens to species-level identification, generating named morphospecies even where formally-described congeners are unavailable. The 44.3% species-level reporting rate I observe in the non-contract portion of DeepData is therefore in a reasonable range for academic CCZ taxonomy, and the much lower 16.4% rate in contract areas is well below what comparable scientific effort would produce.

describes.

A related possibility is that the gap reflects differences in *reporting timing* rather than deliberate underinvestment in taxonomy. Contractors may submit prospectively-mined tract data to DeepData promptly (before taxonomic analysis is complete) in order to satisfy compliance obligations, while non-contract-area samples carry no such pressure and receive more time for refined identifications before upload. If prospectively-mined tract records are submitted earlier in the taxonomic workflow, they would mechanically show lower species-level reporting even if the underlying scientific effort is identical. This is plausible, but it still reflects a difference in how contractors handle data from the two zones—the urgency to demonstrate compliance applies specifically to prospectively-mined tracts because of the permit at stake. That is, even under this interpretation, the asymmetry in reporting precision traces back to the asymmetry in incentives between zones; the same fundamental mechanism the model describes, just operating through the timing of data submission rather than the depth of taxonomic analysis. And if compliance-driven early submission were the primary driver, one would expect contractors to update their records over time; the persistence of the gap suggests that such updates, if they occur, do not close it.

A further possibility is that the gap reflects differences in *sampling volume* rather than reporting incentives. Because contractors sample the tracts they intend to mine far more intensively than the non-contract areas, and because an operation that processes more material might identify each specimen more coarsely, the contract-area coefficient could in principle capture throughput rather than incentive. Adding controls for the number of records collected at each observation’s location and on its research cruise leaves the contract-area coefficient essentially unchanged; if anything the volume terms mostly cut against the concern, since records from more heavily sampled *locations* are reported *more* precisely, not less (only the cruise-level term points the other way, and it leaves the contract-area gap intact). A parallel control for the volume handled by each identifying institution—available only for records carrying identifier metadata—tells the same story (Appendix Table 15). Sampling volume does not account for the zone gap.

Finally, the dependent variable measures taxonomic resolution as *reported in DeepData*, which may not perfectly reflect the underlying taxonomic work. Lower species-level reporting could reflect incomplete database uploads, delays in taxonomic revision, or administrative bottlenecks rather than reduced scientific effort. Rabone, Horton, et al. (2023) document significant data-quality issues in DeepData. These limitations are real, but they would have to affect contract-area and non-contract-area submissions *differentially* to explain the systematic gap I document. Symmetric data-quality issues add noise but cannot generate a systematic bias in one direction.

6.2 Counterfactual: Reporting without Taxonomic Deflation

The model’s second prediction is that contractors could identify organisms more precisely without sampling more, but rationally choose not to—a deflation gap that the welfare analysis then prices as a precision wedge. To quantify the gap, I run a counterfactual exercise that shuts down the $\phi > 0$ mechanism while holding exploration effort and all other covariates fixed. Figure 7 plots the predicted number of specimens that would include a species entry in the absence of taxonomic deflation, out of the 150,104 specimens with complete environmental covariates (the sample on which all four specifications can generate predictions). I generate fitted values from the regressions with the contract-area indicator set to zero for every specimen. That is, I simulate the probability of species reporting as if every specimen were reported with the same taxonomic precision as non-contract-area samples. I use the largest and smallest estimated $|\hat{\beta}|$ to bound the result. As Figure 7 shows, removing the deflation would lead to substantially higher predicted reporting—between 37,372 and 66,467 specimens would report the species of the sampled organism, against just 29,050 in actuality.

The counterfactual also decomposes the paper’s motivating question. Of the 121,054 specimens in the sample that lack a species identification, the share attributable to strategic incentives ranges from 7% (8,322 specimens) under the most conservative coefficient to 31% (37,417 specimens) under the largest. The complementary shares—93% and 69% respectively—would still lack a species identification even absent any deflation, representing the genuine limits of deep-sea taxonomy: undescribed fauna, specimen damage, and scarce expertise that no policy can immediately overcome. Even by the most conservative estimate, strategic incentives account for thousands of missing species identifications in DeepData; under the largest estimate, tens of thousands. The exercise is partial-equilibrium—it holds exploration intensity and all covariates fixed, varying only the reporting precision. The discovery hazard deters sampling as well as precision, as the exploration wedge in equation (9) shows, so removing it would raise exploration too; the counterfactual, which holds exploration fixed, should therefore be interpreted as a lower bound on the full welfare gain from removing the incentive distortion. The headline coefficient is the relevant input for this exercise even though the mechanism section locates much of the gap in between-cruise and between-identifier variation. Those specifications *attribute* the deflation to particular channels (which cruises a contractor sends where, which identifier processes which sample); they do not eliminate it. The contractor’s choice of cruise composition and identifier assignment is itself part of how the firm responds to the permit at stake, so the contractor’s total behavioral response (the relevant object for the social planner) is captured by the headline estimate, not by the within-channel residual.

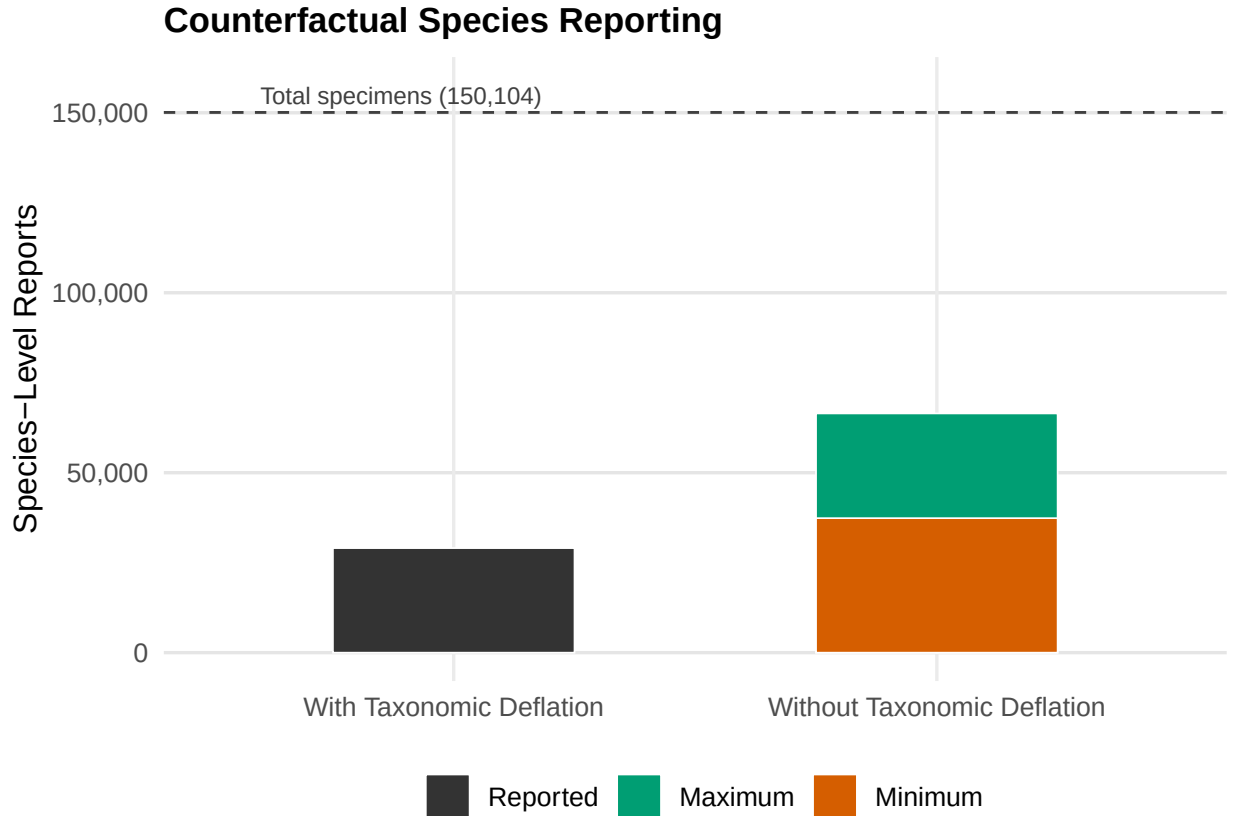


Figure 7: The dark bar plots observed species reports. The vermilion bar plots the counterfactual using the most conservative estimated contract area effect; the green bar extends above it to show the counterfactual using the largest estimated effect. The dashed reference line marks the total number of specimens in the sample: the gap between the counterfactual bars and this line is the share of missing species identifications that taxonomic deflation cannot explain.

6.3 Counterfactual: Exploration Coverage in the APEIs

The precision wedge is only half of the social cost the model identifies. The second is an exploration wedge: because no permit rides on ground the contractor will never mine, its sampling there is pinned by the scientific return alone, and APEIs are barely sampled at all. Figure 2 documented the raw asymmetry: prospectively-mined tracts are sampled roughly 663 times as densely as APEIs, even though the two zones cover comparable area in the CCZ. A second counterfactual asks what that missing exploration would have yielded. The density of prospectively-mined tracts is the natural reference because it is the exploration the ISA’s permit incentive actually elicits—a benchmark grounded in what the regime achieves where it has created an incentive, not an arbitrary target. Scaling APEI sampling to that level is therefore a move toward

closing the exploration wedge, even if not all the way. I take the sampling density of prospectively-mined tracts (0.121 specimens per km²) and apply it to the 1,962,328 km² of APEIs, holding the APEIs' own species-reporting rate (39.9%) fixed so that the exercise varies only sampling volume and not reporting precision. Holding it fixed is what the model licenses: outside the contract areas sampling and precision are independent, so raising exploration there would not by itself move the rate at which recovered organisms are named. Under this scenario the 358 specimens actually recorded inside APEIs become roughly 237,000 (about 87,000 physical sampling events), and the 143 species-level identifications they currently carry become on the order of 95,000—an additional 95,000 named occurrences from the very areas the regime set aside to safeguard biodiversity.

This exercise should be read as a thought experiment, not a forecast. The extrapolation is enormous: it scales 358 actual specimens up by a factor of roughly 663, and the 39.9% reporting rate that anchors the species-report projection is itself estimated from only those 358 specimens—the sampling-noise band alone spans 83,000–107,000 reports. Nor is mined-tract density a normative benchmark: APEIs are set aside precisely so they will not be mined, and there is no reason the socially optimal survey intensity for a reference area should exactly equal that of a commercial tract. Nevertheless, the thought experiment is meant to convey an order of magnitude. The exploration that the contractor regime leaves undone outside the reach of a permit is not a rounding error against the biodiversity record: in fact, the unsampled coverage of the APEIs alone dwarfs the entire identified APEI collection by orders of magnitude.

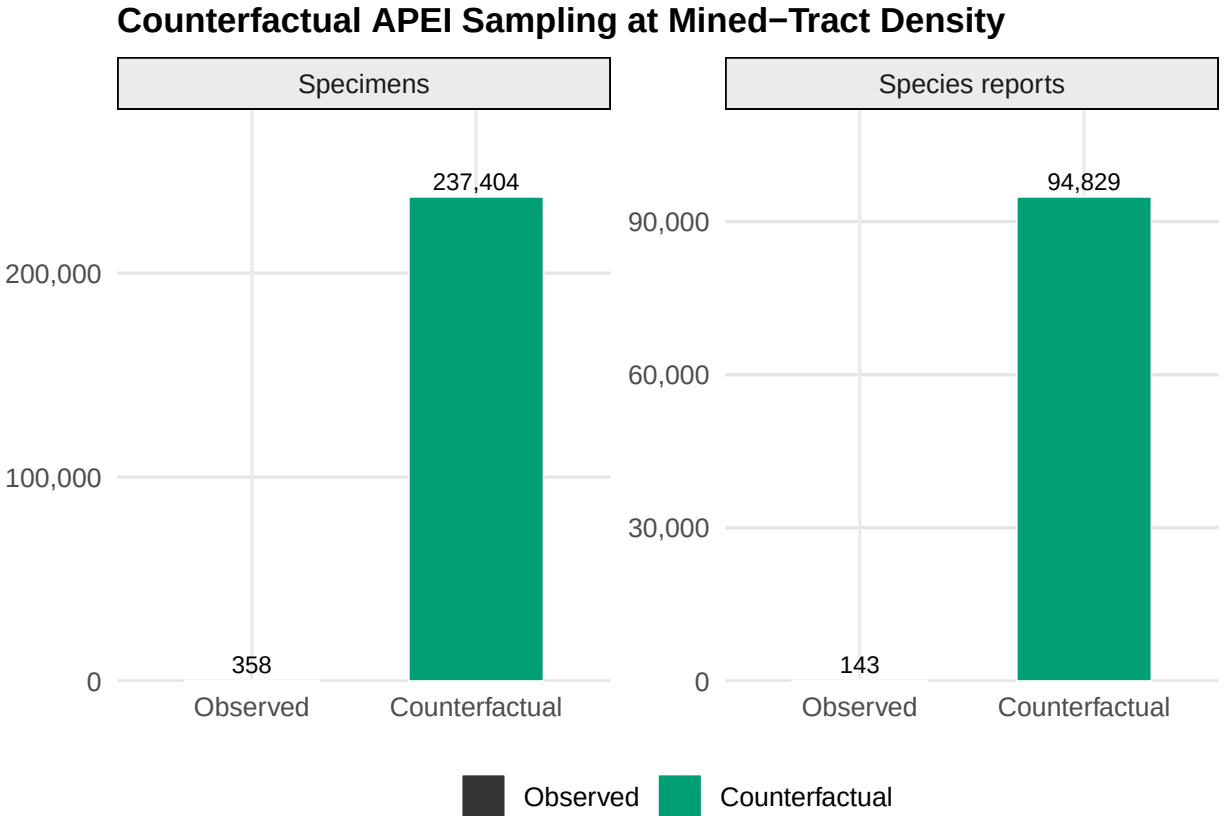


Figure 8: Observed versus counterfactual APEI sampling. The dark bars plot what APEIs actually yield: the specimens recorded inside them and the species-level identifications those specimens carry. The green bars plot the counterfactual in which APEIs are sampled at the per-area density of prospectively-mined tracts, holding the APEIs’ own species-reporting rate fixed so that only sampling volume changes. The counterfactual is a deliberately extreme thought experiment—a several-hundredfold extrapolation from a small observed base—and is meant to bound the order of magnitude of the exploration wedge, not to predict any particular survey outcome.

7 Policy Options

My findings do not suggest that the private requirement to report data is hopelessly flawed. The current arrangement may still be preferable to either a scenario where the seabed is mined without exploration or a scenario where the seabed is explored extensively by a third party prior to mining. Under the first, society loses scientific knowledge, genetic resources, and the opportunity to protect particular discoveries. Under the second, the third party tasked with exploration does not internalize the returns to speed and

may indefinitely delay mining in service of perfect exploration. An ideal system balances the incentive to explore and the incentive to mine against the social optimum. Since the social benefits of exploration are not well-compensated, the ISA’s solution of using the opportunity to mine as a carrot to incentivize exploration may well be the best that can be done, in spite of the perverse incentives I document here. This is true even under the more skeptical view that deep-sea mining may not be economically justified once full environmental costs are included (Sumaila et al. 2023): if mining ultimately does not proceed, the biodiversity data generated under the exploration requirement may prove to be the regime’s most lasting contribution. What the analysis does show is that the regime has a meaningful principal-agent problem. The mining company internalizes the opportunity cost of delaying exploration, but it does not act as a perfect agent of the ISA on the exploration margin, because it does not place equivalent weight on scientific discovery.

If the problem is significant enough to warrant policy changes, the ISA has several options. The exploitation regulations remain under negotiation, with numerous outstanding issues unresolved (Pickens et al. 2024), so there is an opportunity to address deflation incentives before the regulatory framework is finalized. The model’s welfare analysis identifies two distinct wedges—the precision wedge and the exploration wedge—each of which has a menu of possible policy remedies.

The intuitively desirable remedy for the precision wedge is to identify the uncontracted shirking margins and strengthen the reporting requirement against them, reducing the contractor’s ability to exploit the precision wedge. The structure of the Mining Code suggests where to begin: the obligation to identify organisms precisely currently lives only in the Legal and Technical Commission’s non-binding recommendations (International Seabed Authority 2020a), so migrating a minimum-resolution requirement into the binding layer would convert today’s discretionary margin into an enforceable one—the draft exploitation framework already separates legally binding “Standards” from advisory “Guidelines” (International Seabed Authority 2019), a natural vehicle for such a rule. This is easier said than done: many different margins admit deflation. Because my results imply that non-contract areas are reported with greater taxonomic precision than contract areas, more attention to how exploration standards differ between zone types could be fruitful. The ISA should take care to prevent the gaming of any updated rules: a requirement that explicitly compares samples from contract and non-contract areas could create an incentive to similarly underreport from outside areas, while a requirement that explicitly targets reporting of specific taxonomic levels may induce false specificity. This is a Goodhart’s-law problem: once a measure becomes a target, it ceases to be a good measure.

A more targeted approach to the precision wedge would be to decouple discovery from denial. As the model shows, the contractor depresses taxonomic precision below the socially optimal level because precise identification risks triggering conservation review. Several concrete mechanisms could reduce the regulatory

cost of a species discovery. *Safe harbor provisions* would guarantee that a contractor’s mining approval cannot be revoked solely on the basis of species discoveries made during its own survey, removing the direct penalty for precise identification. However, such a guarantee could undermine conservation goals. *Sealed reporting* would require contractors to submit their raw biological specimens and taxonomic data to an independent scientific body, which would conduct or verify the taxonomic analysis separately from the permitting process. The ISA would receive the final taxonomic assessments only after the permitting decision has been made, or would receive them in aggregated or anonymized form that cannot be traced to specific permit applications. This breaks the link between what the contractor discovers and what the regulator uses against it: contractors would have no reason to suppress precise identifications, because those identifications could not influence their own permit outcome. The independent body would still make the full taxonomic record publicly available, preserving the scientific value of the data even if conservation becomes less targeted. These mechanisms address the incentive distortion for mining contractors by reducing the regulatory penalty of a species discovery, but come at the cost of limiting conservation options for the ISA which may be untenable.

A further remedy changes *who* does the science. Rather than relying on the contractor’s incentives, the ISA—or a third party it funds—could provide the biological exploration and identification directly. A funded, non-conflicted party has no permit at stake, so this sidesteps the deflation incentive at its source; in principle it closes both wedges at once, with independent surveys supplying the exploration that contractors underprovide and independent identification supplying the precision they suppress. The taxonomic capacity to identify CCZ fauna at high resolution exists; what is missing is the mandate and funding to apply it. The ISA has begun a limited version of this through its Sustainable Seabed Knowledge Initiative, which funds the description of new deep-sea species (Authority 2022; Jażdżewska and Horton 2026). Because it works on already-collected material, however, SSKI as currently constituted addresses only the precision wedge. The same holds for *independent taxonomic review*, which would assign post-collection identification to third-party specialists through mechanisms outside of the influence of mining contractors, sharpening precision while still leaving sample collection, and its timeliness, to the contractor.

Public provision would also do double duty as enforcement. Independent baseline surveys of non-contract areas—and the APEIs in particular, which Stewart et al. (2026) note remain largely unsurveyed—would both fill the exploration gap and furnish an external benchmark against which contractor submissions could be judged. A regulator holding its own representative sample of CCZ fauna can tell when a contractor’s reported species rate falls implausibly short of what comparable ground yields, turning the reference collection into a yardstick for detecting deflation rather than merely a supplement to the record. Such a program would require careful design, penalizing genuine shirking without punishing firms for the inherent difficulty and

randomness of deep-sea biological sampling.

All of these measures shift cost onto the ISA, and any program that funds new exploration directly reintroduces the tension the contractor model was built to avoid: an explorer who does not internalize the opportunity cost of delaying mining.

To address the exploration wedge, the ISA could incentivize discoveries—directly, by awarding prizes or otherwise allowing monetization of discoveries, or indirectly, by strengthening the relationship between high-effort exploration and mining approval. The instruments that reach both margins at once are those that pay for a *named specimen*—a prize per described species, an advance commitment, access to genetic-resource markets—because the return they create attaches to the product of sampling and precision rather than to either alone. In the model that is the social value s , and making the contractor internalize it raises both effort and precision in non-contract areas, compressing the gap with prospectively-mined tract behavior. An instrument that raised the return to resolution alone would move precision without moving sampling. The result would more closely approximate the economic optimum, in which a single agent internalizes all the relevant costs and benefits for both mining and exploration. But these instruments would be difficult to design appropriately, and could backfire if they end up rewarding contractors for continuing current practice. Nor are the two instruments confined to separate margins *on a tract the contractor holds*: there the regulatory penalty and the scientific return enter both wedges, so either lever reaches exploration and precision alike. Outside the contract areas the division is sharper, since no permit rides on that ground and the regulatory penalty has nothing to bite on: what is left there is the unpriced scientific value alone, which is why the APEI shortfall answers only to instruments that price discovery. Each carries its own monetary costs and legal challenges, so adoption involves clear tradeoffs.

These policy directions connect to the model’s further implications. If deflation intensifies with tract value, monitoring and enforcement resources should be concentrated on the highest-value exploration contracts. The model also cautions against assuming that tighter precautionary standards will on their own reduce deflation: raising the regulatory cost of a discovery depresses precision directly, but it also deters the sampling that would put a discovery at risk, and the net effect is ambiguous—which is a reason to pair any tightening with mechanisms that decouple discovery from denial. And because what precision is worth to the contractor is scientific rather than regulatory, even modest investments in raising that payoff, be it through data prizes, publication incentives, access to genetic-resource markets, or more, act directly on the margin the model identifies.

These policy directions also have parallels in the literature on endangered-species regulation. The ESA creates

similar perverse incentives for private landowners who might harbor listed species (Brown and Shogren 1998; Langpap et al. 2018), and safe-harbor agreements have been shown to mitigate preemptive habitat destruction in that context (Byl 2019). Polasky and Doremus (1998) show that ESA policy gives landowners an incentive *not* to cooperate with species-information collection, and Innes et al. (1998) show that the design of takings compensation substantially shapes landowners' incentives to conserve rather than develop habitat. The deep-sea mining setting differs in one important respect, however: landowners under the ESA already hold property rights over their land, while mining contractors do not own the seabed and are applying to use a common-heritage resource. There is less legal architecture to grant property rights over biological discoveries in the deep sea, which limits the applicability of compensation-based solutions that have proven effective on land. That makes the design of reporting incentives—sealed reporting, independent taxonomic review, discovery prizes—all the more important. The entry into force of the Agreement on Biodiversity Beyond National Jurisdiction (BBNJ) introduces new obligations around environmental impact assessment and marine genetic-resource benefit-sharing that interact with the ISA's existing regime in ways that remain unresolved (Willaert and Soete 2025). If the BBNJ framework creates additional regulatory consequences for species discoveries, it could amplify rather than reduce the incentive to deflate, making it all the more urgent to decouple discovery from denial before these regimes fully intersect.

Each option carries costs and risks. Overly prescriptive requirements could lead to false specificity; excessive reliance on third-party surveys could delay mining without improving biodiversity knowledge. Without action, however, taxonomic deflation threatens the scientific integrity of deep-sea biodiversity assessments.

Several of the remedies above—independent expert review, decoupling discovery from denial, raising the scientific return to precision, and adversarial benchmarking—are not specific to ocean resources or areas beyond national jurisdiction. Each is recognizable in other regulatory domains where the regulated firm produces the information the regulator uses to police it. Independent expert assignment has analogues in blind grading, third-party financial audits, and the assignment of clinical-trial endpoints to independent adjudicators. Decoupling discovery from denial echoes safe-harbor provisions under the Endangered Species Act (Byl 2019) and the broader literature on regulatory dealing under information asymmetry. Raising the scientific return to precision corresponds to publication incentives, data prizes, and market-based instruments for compensating information provision. Adversarial benchmarking is the institutional logic behind environmental NGOs, independent inspector general offices, and academic shadow-monitoring of regulated industries. The deep-sea mining setting allows me to ask which of these mechanisms has empirical purchase on the specific incentive distortions documented above; the broader contribution is that the menu of remedies developed in other domains can be brought to bear on the governance design choices the ISA is currently making.

8 Conclusion

The motivating question of this paper has a partial answer: the imprecision of deep-sea mining data is not fully explained by scientific difficulty. The difficulty is real, but it is symmetric across zones. The asymmetry in species reporting that I identify is the signature of taxonomic deflation. The counterfactual decomposition implies that even by the most conservative estimate, strategic incentives account for roughly 7% of the missing species identifications in DeepData—around 8,000 specimens under that bound, and as many as 37,000 under the largest—all of which would carry a species name if discovery did not threaten the permit. The same logic has a second, larger edge. Because no permit rides on the non-contract areas, they go almost entirely unsurveyed. Specifically, the APEIs are sampled roughly 663 times less densely than prospectively-mined tracts, and had they been explored at mined-tract density they would have yielded on the order of 95,000 additional species-level records rather than the 143 they now contain. The underlying reality is that no single entity has fully aligned incentives to balance exploration and exploitation of the deep sea, and the ISA’s operative solution—using the prospect of mining as a carrot for biodiversity reporting—inevitably creates the precision and exploration wedges I document. Whether that wedge is small enough to tolerate or large enough to fix is a choice the exploitation regulations now under negotiation will encode. Those choices will determine whether the deep-sea biodiversity record reflects what the data could show or only what mining contractors find it convenient to report.

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Appendix

The appendix is organized into three sections. **Extra Tables** and **Extra Figures** present material referenced from the main text that did not meet the bar for in-line inclusion. **Additional Analyses** presents exercises that are tangential to the central argument but that may be of interest for evaluating specific concerns.

Extra Tables

Table 5: Summary statistics for biological specimen records in the DeepData database. Contract Areas are contractor-allocated exploration tracts; APEIs are Areas of Particular Environmental Interest; Outside Areas are samples outside both zone types. Species reporting rate is the proportion of records that include a species-level identification.

Zone	Specimen Records	Contractors	Year Range	Species Reporting Rate
Contract Areas	149,795	12	2004–2021	16.4%
APEIs	366	6	2015–2018	41.5%
Outside Areas	10,806	7	2010–2021	44.4%
Total	160,967	12	2004–2021	18.3%

Table 6: Linear probability model estimates of species-level reporting. Dependent variable: indicator for species-level taxonomic identification. Comparison group is non-contract areas (APEIs and outside areas). Robust standard errors in parentheses.

	Contract Area	+ Contractor FE	+ Year FE	+ Environmental Controls
Intercept	0.443*** (0.005)	0.529*** (0.026)	1.233*** (0.037)	−23.842*** (1.830)
Contract Area	−0.279*** (0.005)	−0.192*** (0.005)	−0.091*** (0.006)	−0.060*** (0.006)
Depth (m)				0.000*** (0.000)
Sea Surface Temp.				0.320*** (0.022)
Surface Chlorophyll				−5.603*** (0.507)
Dissolved Oxygen				3.496*** (0.274)
Contractor FE	No	Yes	Yes	Yes
Year FE	No	No	Yes	Yes
Environmental Controls	No	No	No	Yes
Num.Obs.	160 967	160 967	160 967	150 104
R2	0.034	0.123	0.213	0.211

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 7: Taxonomy fixed effects robustness. Dependent variable: species-level identification. Comparison group is non-contract areas (APEIs and outside areas). All specifications include contractor, year, and environmental controls. Each specification restricts the sample to observations with a valid identification at the relevant taxonomic level. Robust standard errors in parentheses.

	Baseline	+ Phylum FE	+ Class FE	+ Order FE	+ Family FE	+ Genus FE
Intercept	−23.842*** (1.830)	−13.079*** (1.629)	−6.905*** (1.630)	−1.407 (1.933)	−4.165** (1.926)	−8.189*** (1.444)
Contract Area	−0.060*** (0.006)	−0.051*** (0.006)	−0.051*** (0.005)	−0.043*** (0.005)	−0.049*** (0.004)	−0.068*** (0.004)
Contractor FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Environmental Controls	Yes	Yes	Yes	Yes	Yes	Yes
Phylum FE	No	Yes	No	No	No	No
Class FE	No	No	Yes	No	No	No
Order FE	No	No	No	Yes	No	No
Family FE	No	No	No	No	Yes	No
Genus FE	No	No	No	No	No	Yes
Num.Obs.	150 104	139 331	131 863	85 873	73 042	63 989
R2	0.211	0.330	0.560	0.662	0.768	0.823

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Sample Source Composition

Each biological specimen record in DeepData is reported by a specific mining contractor and is associated with a specific physical location. Using the official ISA contract polygon shapefiles—downloaded from the ISA’s Maps repository at isa.org.jm/exploration-contracts/maps/—I match each observation to the contract polygon (if any) within whose boundary it falls. Combined with the identity of the *sampling* contractor (extracted from the `area_key` field), this yields four mutually exclusive observation types:

- **Own tract:** the observation falls inside the sampling contractor’s own PMN polygon—this is the regulatory regime where a permit is at stake.
- **Other tract:** the observation falls inside *another* contractor’s PMN polygon—e.g., a German (BGR) sample collected within Singapore’s (OMS) tract during a joint or transit cruise. The sampling contractor has no permit-related incentive at this location.
- **APEI:** the observation falls inside an Area of Particular Environmental Interest, where mining is prohibited.
- **Outside all:** the observation falls in international waters outside any contract polygon and outside any APEI.

This decomposition is helpful both descriptively, as it characterizes where the dataset’s biological samples

actually come from, and methodologically, since the four types differ in their regulatory exposure. Table 8 reports counts and species-level reporting rates for each type. Roughly 93% of CCZ specimens are collected in the sampling contractor’s own tract; 6% are collected by one contractor inside another’s tract; and APEIs and truly-outside observations together make up less than 1%. The pattern in the species-reporting column is striking. Specimens collected by a contractor within their own tract are reported with species-level identification only 17.6% of the time. The same contractors, when they sample inside *another* contractor’s tract, report at the species level 46.7% of the time. In APEIs, where mining is prohibited and there is no commercial interest at stake, contractor-collected specimens reach 39.9% species-level reporting. The sharp gap between “own tract” and “other tract” samples that share the same physical region motivates the within-sampler and within-polygon robustness exercises reported in Tables 1 and 2 of the main text.

Table 8: Composition of biological observations by spatial source. "Own tract" = inside the sampling contractor’s own PMN polygon; "Other tract" = inside another contractor’s PMN polygon; "APEI" = inside an Area of Particular Environmental Interest; "Outside all" = outside any PMN polygon or APEI. Spatial classification uses official ISA shapefiles.

Sample source	Observations	Share of total	Species-level reporting
Own tract	140,255	93.2%	17.6%
Other tract	9,045	6.0%	46.7%
APEI	358	0.2%	39.9%
Outside all	801	0.5%	48.6%

Stopping-Level Analysis

The main paper’s headline outcome is binary: whether a record includes a species-level identification. This collapses the taxonomic hierarchy into a single threshold (species reached vs. not), which obscures *where* an incomplete identification stops. Some records get to genus and stop; others stop at family, or order, or class. This subsection presents a richer recoding that captures the depth of each identification.

For each record I construct a *stopping level* taking integer values 0–7: 0 if no taxonomic field is populated, 1 if only kingdom, 2 if at most phylum, 3 if at most class, 4 if at most order, 5 if at most family, 6 if at most genus, and 7 if species is reached. Table 9 reports the distribution of stopping levels by zone, and Table 10 reports a linear regression of stopping level on the contract-area indicator with the standard control set.

Contract-area records stop, on average, about two-thirds of a taxonomic level shallower than non-contract-area records, even after controlling for contractor, year, and environmental covariates. The distributional pattern in Table 9 is consistent with the headline result of Table 6 but adds texture: the gap is not a binary “species reached vs. not” — it is a graduated tendency for contract-area records to stop earlier in the taxonomic

Table 9: Distribution of taxonomic stopping levels by zone (% within zone). Each record is assigned the deepest taxonomic level it reaches (0 = no kingdom reported, 7 = species reported). Contract-area records concentrate around class and genus; non-contract records concentrate at species.

Stopping level	Contract	Non-contract
0 — None reported	6.0	0.1
1 — Kingdom	1.0	0.0
2 — Phylum	4.3	2.1
3 — Class	31.4	13.3
4 — Order	10.3	21.8
5 — Family	5.7	7.8
6 — Genus	24.8	10.6
7 — Species	16.4	44.3

Table 10: Linear regression of stopping level on contract-area indicator with the standard control set. The coefficient is interpretable as the average difference in taxonomic depth (out of 7 possible levels) between contract-area and non-contract-area records, conditional on contractor, year, and environmental covariates. Robust (HC1) standard errors in parentheses.

	Stopping level
Contract Area	−0.673*** (0.022)
Contractor FE	Yes
Year FE	Yes
Environmental Controls	Yes
Num.Obs.	150 104
R2	0.252

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

hierarchy.

Identifier-FE Selection-Bias Check

The identifier-fixed-effects specifications of Table 3 shrink the contract-area coefficient from -0.060 (Spec 4 of Table 6) to approximately zero. Two competing readings are consistent with that shrinkage. One is that identifier identity absorbs the deflation channel: the same person produces statistically identical work across zones, and the gap reflects which identifier processes which sample. The other is selection bias: the identifier-present sub-sample (135,381 of 150,104 analysis records) is not a random draw from the full sample, and the headline gap might already be small on that sub-sample even before identifier FE absorption.

Table 11 disentangles the two by re-running the standard contractor + year + environmental controls specification on each identifier-FE sub-sample *without* the identifier fixed effects. If the identifier FE genuinely absorbs the effect, these “no-FE” baselines should still recover something close to the headline -0.06 ; if it is

sample selection, the no-FE baselines will already be close to zero.

Table 11: Identifier-FE selection-bias check. Column (1) re-runs the standard contractor + year + environmental controls specification on the identifier-present sub-sample (the same observations used in column (3) of Table \@ref(tab:reg-table-mechanisms)), but without identifier fixed effects. Column (2) does the same on the balanced sub-sample used in column (4) of Table \@ref(tab:reg-table-mechanisms). The contract-area coefficient under no-FE on these sub-samples isolates the share of the identifier-FE attenuation that comes from sample selection rather than from FE absorption.

	ID sample, no FE	Balanced ID sample, no FE
Contract Area	−0.030*** (0.007)	0.037*** (0.007)
Num.Obs.	135 381	21 686
Contractor FE	Yes	Yes
NA	NA	NA
Year FE	Yes	Yes
Environmental Controls	Yes	Yes
Identifier FE	No	No
R2	0.241	0.232

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The no-FE contract-area coefficient on the identifier-present sub-sample is -0.030 (SE 0.007), and on the balanced sub-sample is 0.037 (SE 0.007). The interpretation depends on whether these no-FE estimates are close to the headline -0.060 or close to the identifier-FE estimates near zero: the former would confirm that identifier identity is genuinely absorbing the deflation channel, while the latter would imply that the identifier-FE attenuation is largely an artifact of the sub-sample over which it is estimated. The result on the balanced sub-sample is especially diagnostic because that sub-sample restricts to identifiers who appear in both zones—a population that is, by construction, more comparable across zones than the full identifier-present sample.

Administrative Effort by Zone

The cross-zone asymmetry in field-collection effort flagged in Section 5 extends to compliance-metadata fields used to document the taxonomic identification process. Table 12 reports the field-completion rate for `taxonomist_yes_no`, the field that records whether a taxonomist was involved in identifying a specimen. Contractor data for permit-relevant zones report this field at a rate roughly 30 times that of guest-collected or APEI records. The same pattern obtains for the `identification_method`, `identification_status`, and `identification_date` fields. This is consistent with substantially more administrative effort being invested in the documentation of records that bear on the contractor’s own permit.¹⁴

¹⁴A competing reading of this asymmetry is that contractor submissions for their own tracts are routed through the ISA-mandated reporting template, which includes these compliance-metadata fields, while samples from outside the contractor’s tract—collected on guest visits or contributed by independent expeditions—may not be processed through the same template

Table 12: Field-completion rates for four compliance-metadata fields by zone. Each cell reports the percentage of records in that zone with a non-missing, non-sentinel value in the named field. Contract-area records are filled in at substantially higher rates across all four fields.

zone	Records	taxonomist_yes_no	identification_method	identification_status	identification_date
APEI	350	0.0%	28.3%	8.0%	8.0%
Contract	149,795	72.7%	67.9%	73.0%	73.0%
Outside	10,822	2.1%	9.3%	5.6%	5.6%

and therefore arrive at the database without those fields populated. Under that reading, the differential reflects which submission pipeline a sample passes through rather than discretionary effort allocation. The two readings are not mutually exclusive: contractors choose how thoroughly to fill mandatory fields, and the choice of whether non-contract samples are submitted through a documented pipeline at all is itself an effort choice. The pattern is reported here as a contextual observation.

Extra Figures

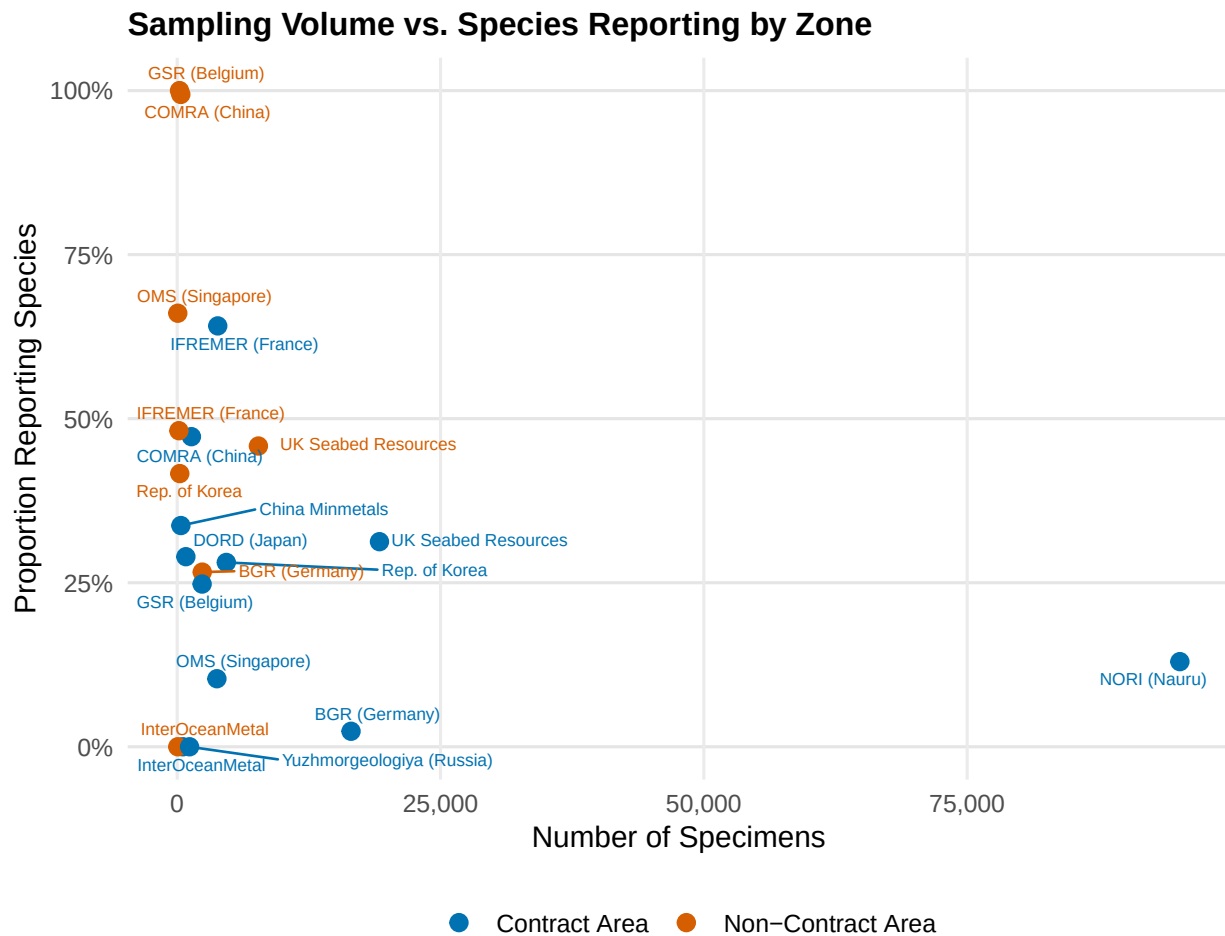


Figure 9: Sampling volume (number of biological specimens) versus species reporting rate by contractor and zone type. Each point represents one contractor's submissions from either contract areas (blue) or non-contract areas (vermillion). The negative cross-zone correlation—more samples but lower species reporting in contract areas—is consistent with the model's twin predictions and difficult to reconcile with a resource-constraints explanation.

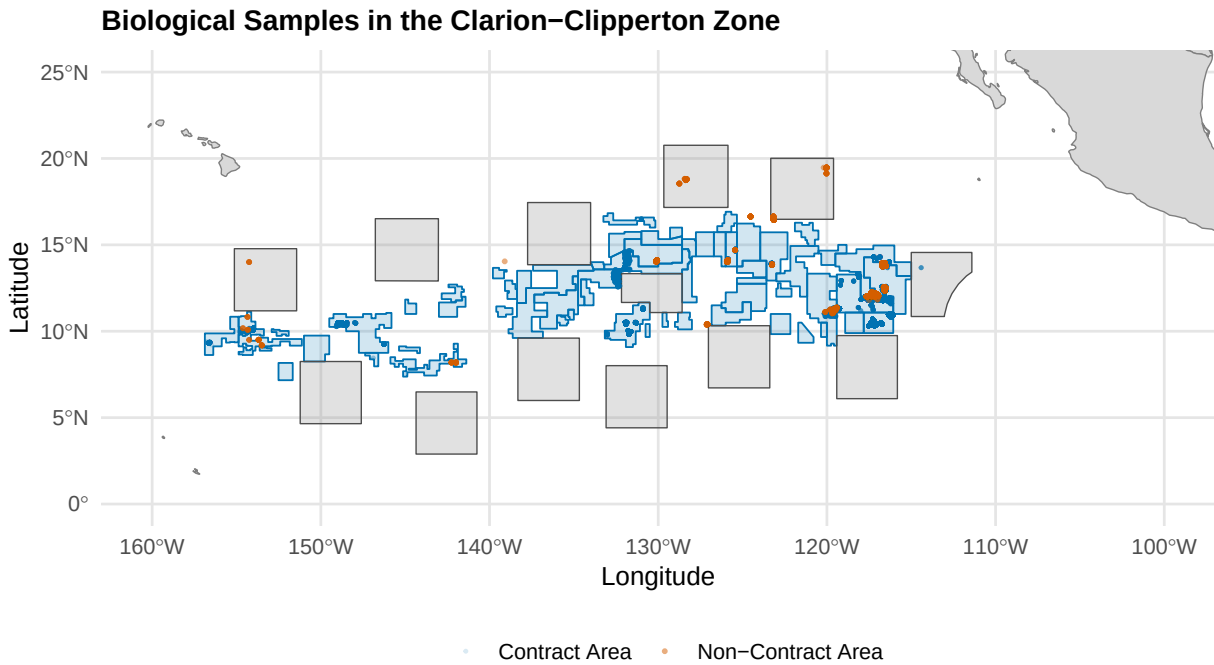


Figure 10: Sample locations overlaid on official ISA contract area polygons (blue) and APEIs (grey). Each point is one biological sample submitted to DeepData. Source for polygons: International Seabed Authority.

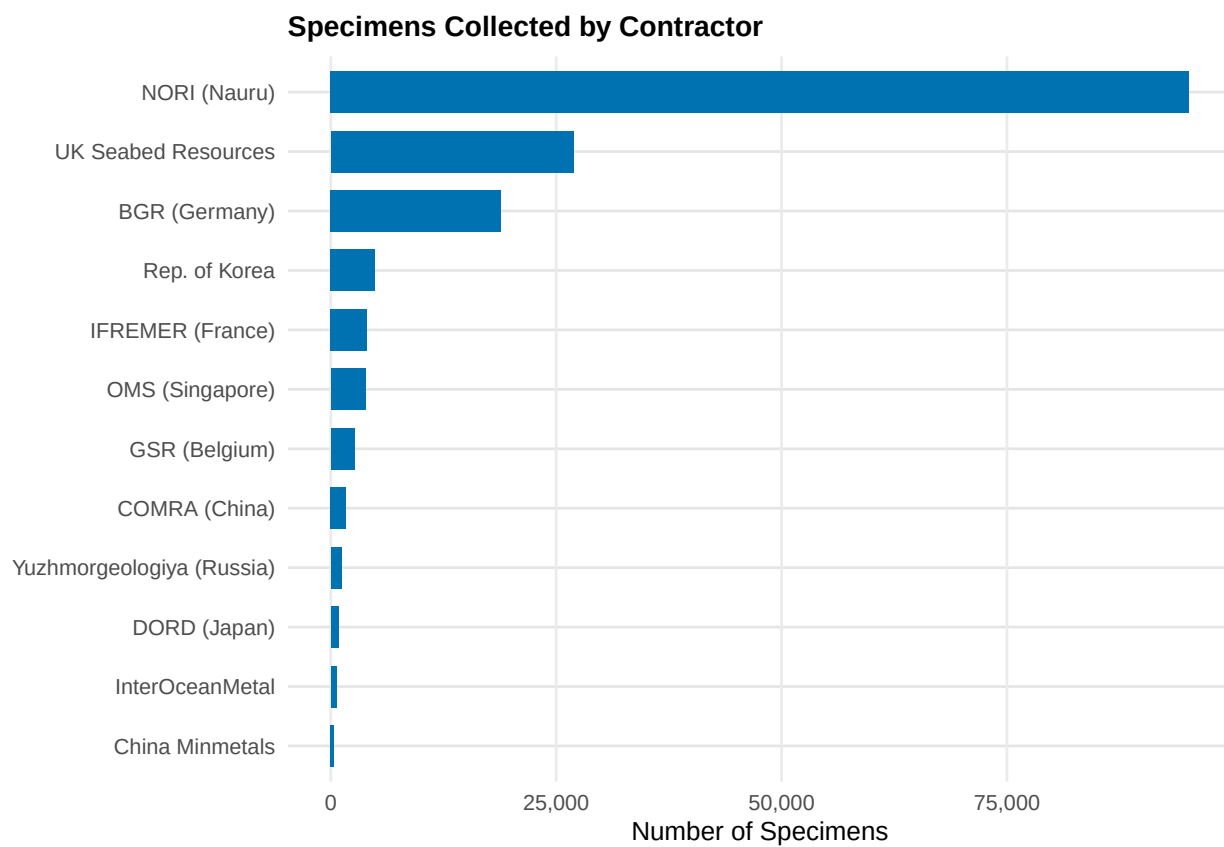


Figure 11: Number of biological specimens collected by each contractor in the CCZ.

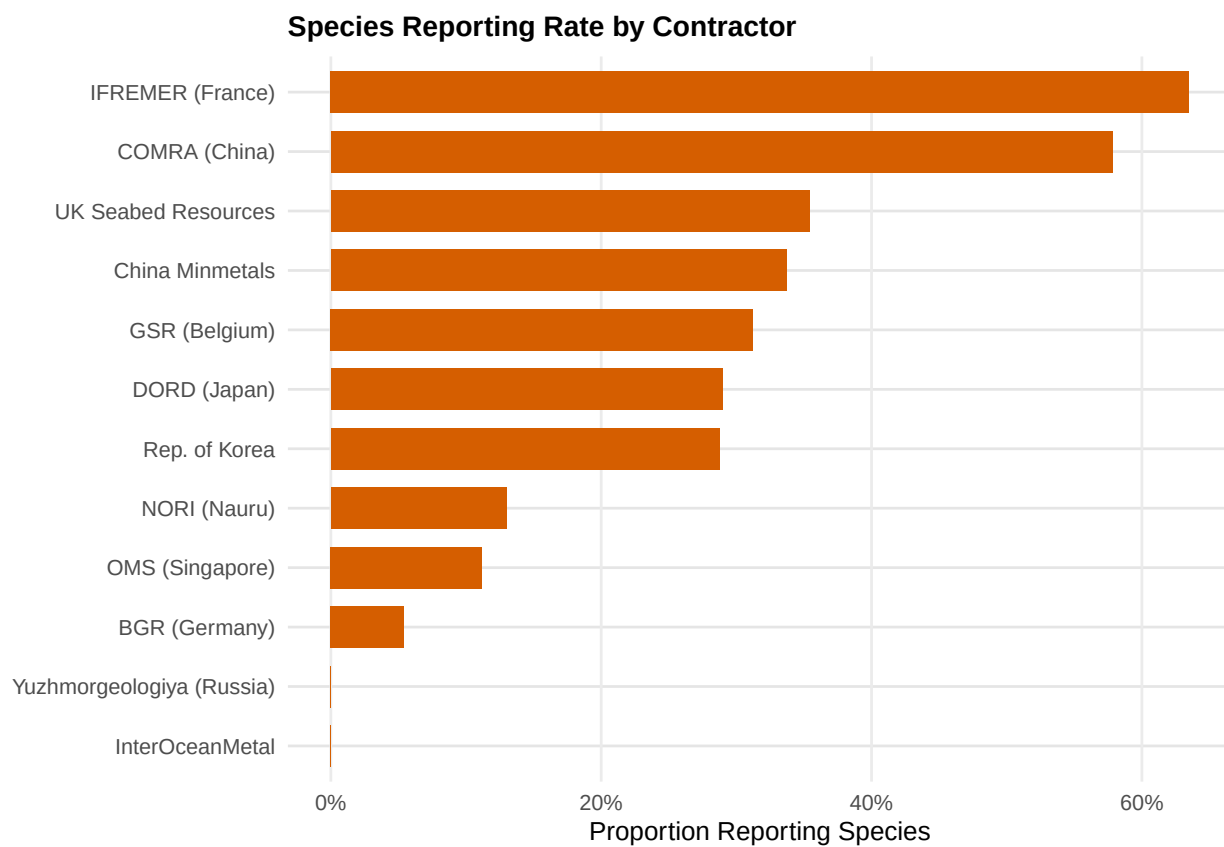


Figure 12: Proportion of observations reporting species-level identification by contractor.

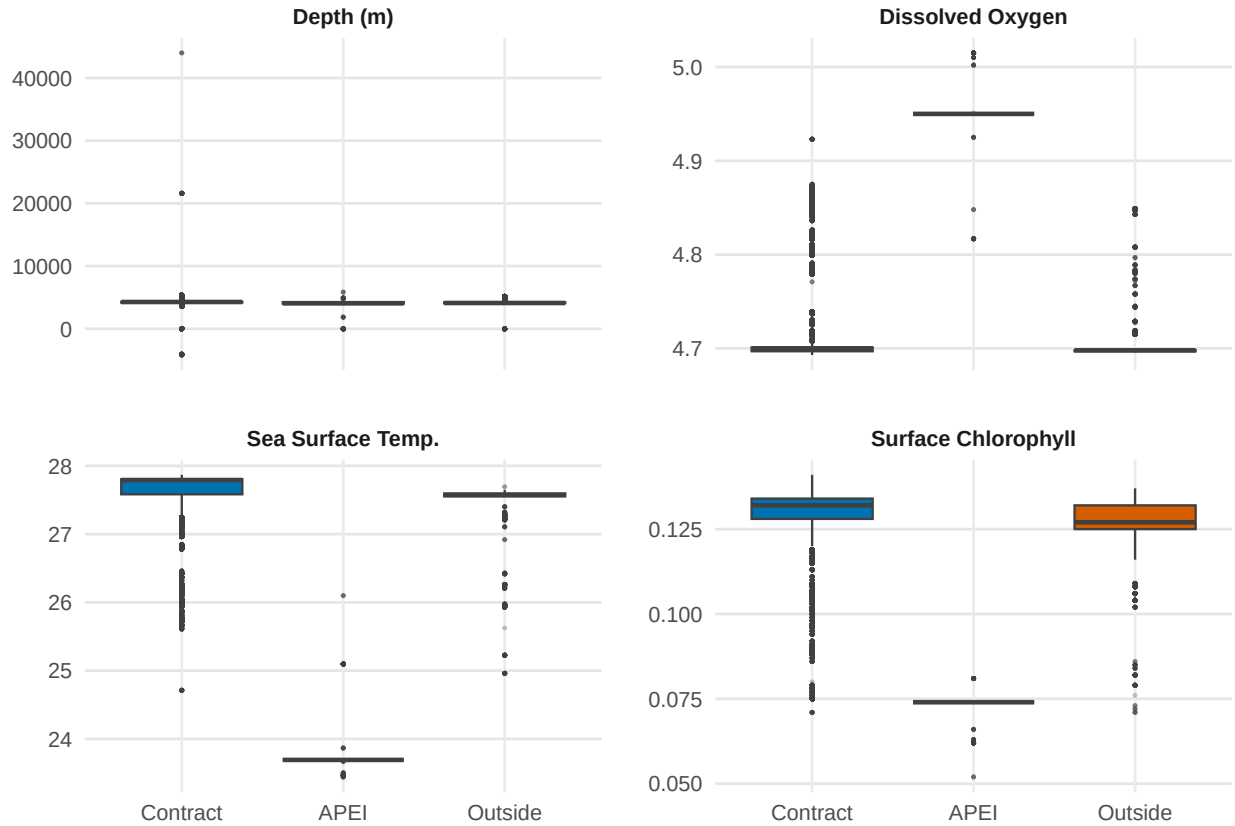


Figure 13: Distributions of the four environmental covariates used in the main analysis, by zone type. Bio-ORACLE values are matched to each sample’s longitude and latitude. The contract-area and APEI distributions overlap heavily, consistent with the ISA’s representativity criterion for APEI siting; outside areas show somewhat broader spread on most variables. The remaining marginal heterogeneity is absorbed by the environmental controls in the main regression.

Additional Analyses

APEI-Only Comparison

Tables 13 and 14 replicate the main regression analyses restricting the comparison group to Areas of Particular Environmental Interest only, excluding outside areas. The APEI-only sample is the cleanest comparison: APEIs are explicitly designated non-contract areas with no mining permit at stake, and all APEI samples in the dataset are contractor-submitted, holding fixed differences in reporting infrastructure and scientific methodology. The sample is considerably smaller (APEIs contribute only 374 records), but the directional results are consistent with the broader analysis.

Table 13: APEI-only comparison. Linear probability model estimates restricting the sample to contract areas and APEIs. Dependent variable: indicator for species-level taxonomic identification. Robust standard errors in parentheses.

	APEI	+ Contractor FE	+ Year FE	+ Environmental Controls
Intercept	0.164*** (0.001)	0.337*** (0.025)	1.077*** (0.034)	-46.931*** (2.099)
APEI	0.243*** (0.025)	0.088*** (0.026)	0.063* (0.034)	0.822*** (0.088)
Depth (m)				0.000*** (0.000)
Sea Surface Temp.				0.762*** (0.034)
Surface Chlorophyll				-10.259*** (0.516)
Dissolved Oxygen				5.863*** (0.288)
Contractor FE	No	Yes	Yes	Yes
Year FE	No	No	Yes	Yes
Environmental Controls	No	No	No	Yes
Num.Obs.	150 161	150 161	150 161	139 675
R2	0.001	0.099	0.219	0.220

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 14: APEI-only taxonomy fixed effects robustness. Sample restricted to contract areas and APEIs. All specifications include contractor, year, and environmental controls. Robust standard errors in parentheses.

	Baseline	+ Phylum FE	+ Class FE	+ Order FE	+ Family FE	+ Genus FE
Intercept	-46.931*** (2.099)	-30.766*** (1.899)	-22.993*** (2.020)	-18.771*** (2.324)	-12.226*** (2.421)	-5.690*** (1.885)
APEI	0.822*** (0.088)	0.468*** (0.067)	0.777*** (0.084)	0.853*** (0.089)	0.663*** (0.104)	0.421*** (0.078)
Contractor FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Environmental Controls	Yes	Yes	Yes	Yes	Yes	Yes
Phylum FE	No	Yes	No	No	No	No
Class FE	No	No	Yes	No	No	No
Order FE	No	No	No	Yes	No	No
Family FE	No	No	No	No	Yes	No
Genus FE	No	No	No	No	No	Yes
Num.Obs.	139 675	128 942	121 777	77 584	67 253	58 984
R2	0.220	0.339	0.576	0.657	0.759	0.825

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The pattern in the species-reporting column is striking. Specimens collected by a contractor within their own tract are reported with species-level identification only 17.6% of the time. The same contractors, when they sample inside *another* contractor's tract, report at the species level 46.7% of the time. In APEIs, where

mining is prohibited and there is no commercial interest at stake, contractor-collected specimens reach 39.9% species-level reporting. The sharp gap between “own tract” and “other tract” samples that share the same physical region motivates the within-sampler and within-polygon robustness exercises reported in Tables 1 and 2 of the main text.

Robustness to Sample Volume

Because contract areas are sampled more intensively than non-contract areas, a natural concern is that the contract-area coefficient reflects *sampling volume*, the sheer throughput of specimens, rather than a reporting incentive. If operations that process more material identify each specimen more coarsely, volume alone could generate the gap. Table 15 adds, to the full specification (column 4 of Table 6), the logged number of records sharing each observation’s *location* (its latitude and longitude), its *research cruise*, and its *identification institution*—the laboratory credited with the taxonomy.

Table 15: Robustness to sample volume. Linear probability models adding the logged number of records at each observation’s location, research cruise, and identification institution to the full specification. Dependent variable: indicator for species-level identification. The two institution-volume columns are restricted to records that carry an identifier institution. Robust standard errors in parentheses.

	Baseline	+ Location vol.	+ Cruise vol.	Baseline (ID lab)	+ Institution vol.
Contract area	−0.060*** (0.006)	−0.053*** (0.006)	−0.064*** (0.006)	−0.013* (0.007)	−0.015** (0.007)
log(Location volume)		0.063*** (0.000)			
log(Cruise volume)			−0.044*** (0.003)		
log(Institution volume)					0.101*** (0.003)
Contractor FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Environmental Controls	Yes	Yes	Yes	Yes	Yes
Num.Obs.	150 104	150 104	150 104	127 323	127 323
R2	0.211	0.312	0.212	0.288	0.302

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The location and cruise controls (columns 1–3) are estimated on the full sample, and the contract-area coefficient barely moves. The volume terms themselves run in opposite directions: records from more heavily sampled *locations*, and those handled by higher-throughput *institutions*, are reported with *higher* precision, while larger *cruises* are reported with somewhat lower precision—the only level at which the “high-volume, low-precision” intuition holds.

The institution-volume control (columns 4–5) can be computed only for the 86% of records that carry an identifier institution, and on that subsample the contract-area coefficient is smaller before any volume control

is added (column 4), which the volume control then leaves unchanged. That smaller estimate reflects the composition of which records name an institution, not sample volume—the role of the identifying institution is taken up directly in the mechanism analysis of Section 5.6. Sample volume, at every level I can measure it, does not account for the contract-area gap.